Adjoint-Faithful FNO Project Tracker

MITgcm baroclinic double-gyre benchmark

Status date: 29 July 2026 Canonical plan: `AF_FNO_Project_Plan.tex`

Current position. S0–S2 and the shared trajectory dataset are complete, and A0, A, and B remain frozen controls. Model C's training-only C1b, 320-epoch C1c, and width-64 capacity tests all completed but were scientifically rejected because temperature or SSH did not beat persistence. GPU audit job 287581 checked every evaluated epoch and the exact late-checkpoint objective gradients; it found an already substantial, SSH-dominated increment gradient and supported late optimization stabilization rather than another capacity increase. Bounded GPU job 287583 retained loss v1 and reduced the learning rate from 0.0005 to 0.0001 after epoch 240; this schedule passed every 96-record memorization and reload criterion at epoch 315. The separately versioned 0.0025 increment-weight loss v2 was rejected.

The predeclared validation-only search is now complete. GPU jobs 287604/288466/289379/289915 executed all four from-scratch $10 \to 5 \to 3 \to 2 \to 1$ rounds and selected $(24, 16)$, width 64, four layers, and 10% padding. Neither the capacity nor wall leakage trigger fired, so no six-layer or 20%-padding branch was opened. Final-seed array 290382 completed all three runs with exact three-step reload, but their SSH validation ratios were 1.00014/1.00383/1.03818; U, V, and temperature beat persistence for every seed. The immutable decision is therefore `scientifically_rejected_three_seed_gate`, not an operational failure. Model C is not frozen and inference is not authorized. Validation was used only under the frozen search contract; inference, intermediate-wind, response, and adjoint data remain sealed. Post-search data-adequacy job 290415 subsequently authorized a non-destructive two-times S0–S2 expansion while localizing a separate low-wind S1 training failure.

That expansion is now complete. MITgcm array 290439 produced all three exact ten-year continuations normally in 15m10s–15m31s. Jobs 290443/290444 built and independently validated the 11 GB, $3 \times 7200 \times 46 \times 62 \times 62$ `trajectories_v2` store, including a bitwise-identical v1 prefix and raw-MDS extension spot checks. Training-only coverage job 290446 then passed its predeclared two-times slow-field target: temperature/SSH state effective multipliers are 2.142/2.039 and increment multipliers are 2.106/2.253. Fresh v2 validation and inference remain unread. Successor contract `model_c_successor_training_v1` freezes an exact data-only control, an isolated width-64 $4C$ Channel-MLP ablation, and only then the width-128/256-lift/256-projection/$4C$ candidate. Both tasks in GPU array 290448 completed normally but failed only S1 SSH at 1.103435 and 1.003573 times persistence, respectively. Job 290597 completed width 128 normally and passed every training gate, including S1 SSH at 0.762683. Corrected validation contract SHA-256 `56bb6ec59799...0cf27b` is frozen. Split-1 array 290673 completed both missing replicas normally; all three seeds pass every training gate at worst regime/group ratios 0.762683/0.732270/0.839507. Fresh-v2 validation job 290738 completed operationally but rejected C scientifically: surface speed passes, while SST/surface-pressure RMSE-AUC ratios to persistence are 2.313/2.198 after good day-10 skill. Frozen training-only rollout-diagnosis job 291102 completed with exit zero and reproduced the same drift for every seed. Mean training day-10/AUC/day-90 ratios are 0.809/2.021/4.221 for SST and 0.552/1.867/3.873 for surface pressure, classifying an objective/checkpoint-gate mismatch. A baseline-visible Figure-4 companion now shows that SST loses persistence and climatology at day 20, while surface P/ρ loses persistence at day 30 and climatology at day 70. Exact-replay job 291164 then completed in 1h19m14s and reproduced step 14,880 bit for bit with zero training-history difference. None of the six late checkpoints passes the frozen long-horizon gate; diagnostic best step 14,400 still reaches day-90 SST ratios 3.669/5.513 to persistence/climatology and surface-pressure ratios 3.800/2.247. The applied classification is `objective_correction_required`.

Pushforward job 291612 completed normally in 20m37s. Its final step 1,920 is best by every frozen priority and reloads exactly. Worst primary AUC improves from 2.366 to 1.692, worst slow-field lead ratio from 5.513 to 3.994, and ten-day worst regime/group ratio from 0.852 to 0.610, but the complete gate remains failed. Day-90 SST ratios are 2.659/3.994 and surface-P/ρ ratios are 3.222/1.905 to persistence/climatology. Replication and validation are not authorized.

Duration job 291616 completed normally in 51m31s and reproduced step 1,920 bitwise before adding all 3,840 low-rate steps. None of eight checkpoints passes. Diagnostic-best step 5,760 improves worst primary AUC/slow-field lead ratios to 1.597/3.609, but day-90 SST remains 2.402/3.609 and surface-P/ρ 2.922/1.727 times persistence/climatology. The contract therefore rejects undertraining as a sufficient explanation and applies the predeclared terminal-only-structure diagnosis.

Corrected truncated-unroll contract SHA-256 `0c5aabadbe51...92e3c` changed only the correction gradient graph. Initial job 291769 stopped before training because its validator used the wrong duration-checkpoint step-field name. Version 2 checked the actual schema without changing the objective, and V100 job 291778 completed normally in 20m01s. All four checkpoints fail. Diagnostic-best truncated step 1,920 reloads nine calls bitwise, but worst primary AUC/slow-lead ratios 1.933/3.701 are worse than the duration source at 1.597/3.609. No replica, fresh-validation, inference, or Model-D job is submitted; the next action is no longer another unconstrained temporal variant.

Frozen bias/projection job 292064 completed normally in 7m09s. Static 9ϵ explains 33.43%/27.21% of pooled SST/SSH day-90 error energy, while basin mean plus five increment EOFs explain only 4.57%/1.73%. Bias plus constraints improves the worst slow-field AUC ratio from 1.986 to 1.603 (19.25%), just below the frozen 20% threshold; conservation alone slightly worsens it to 1.996. Apply the feedback-amplification branch, retain the exact linear constraints and a modest bias term, and prioritize forecast-age supervision.

Loss-v3 job 292293 completed normally in 27m46s. Selected diagnostic step 960 is reload-exact and keeps the projection residuals below 1.17×10^{-10}. All ten-day groups pass, but the long gate does not: worst primary AUC improves from 1.597 to 1.462 while worst slow-field lead changes from 3.609 to 3.634. SST RMSE-AUC is 1.119/1.462 and surface-P/ρ is 1.364/0.615 times persistence/climatology. At the mandatory day-10/day-30 adjoint horizons, however, all Bire-primary fields and streamfunction beat both persistence and climatology. The next action is to freeze a separate prospective 10–30-day adjoint-readiness gate with damped persistence and anomaly-scale drift before any new validation read. The two-state/two-output model remains a bounded parallel long-rollout arm; no automatic training variant is authorized. All later archives remain sealed.

The Bire-archive comparison now motivates one higher-priority representation test. Contract `config/model_c_anomaly_direct_v1.json`, SHA-256 `fa38b41578f5...8acea`, freezes pooled split-1 pointwise temporal mean/standard-deviation normalization and direct future-anomaly-state prediction. The scale floor is each channel's wet-cell fifth percentile plus a 10^{-6} guard, affecting exactly 5% of wet points per channel. Width 128, four layers, modes $(24, 16)$, 10% padding, the $4C$ Channel MLP, loss v1, optimizer, data order, and 540-record training-only gate remain fixed. Eight focused/figure tests, lint, shell validation, immutable source and normalizer hashes, and a production-grid three-step forward/backward check pass. V100 job 303447 completed normally in 1h29m32s and selected step 14,400 using split 1 only. The training-only gate passes at worst primary AUC/slow-field-lead ratios 0.394/0.519. On the fixed S2 characterization, speed/SST/surface-P/ρ 10–90-day AUC ratios to persistence are 0.222/0.206/0.601 and day-200 RMSEs remain bounded at 0.00212 $\mathrm{m\,s^{-1}}$, 0.0288 °C, and 0.0552 $\mathrm{m^2\,s^{-2}}$. The required Figure 4 and all five companion plots are complete. Replication array 303640 completed both seed-only tasks normally, and all three seeds pass the frozen AUC, boundedness, reload, and seal checks. Ranking scores 0.30227/0.24717/0.23184 select the predeclared median seed 20260724 at step 13,440; its day-200 speed/SST/surface-P/ρ RMSE is 0.00242/0.0262/0.0223. Summary/CSV/manifest/plot are complete under `outputs/af_fno/C/anomaly_direct_replication_v1/`.

The held-inference contract is frozen before split 3 is opened:

`config/model_c_anomaly_direct_forward_v1.json`
SHA-256 `d11029fce3aa...c2f6c`.

It fixes 45 fresh-block trajectories through day 360, the median checkpoint, persistence, regime climatology, damped persistence, same-start A0, the separate day-10/day-30 adjoint-readiness decision, and all long-forward physical gates. V100 job 303993 completed normally in 3m00s. The 10/30-day decision passes, all primary 10–90-day AUCs beat every baseline, the one-year rollout is finite and bounded at normalized maximum 15.81 with negligible group-mean drift, and the wind-slope ratio is 1.180. The combined gate fails only because day-360 mid/bottom-P/ρ modewise spectral ratios 22.29/9.63 exceed four; the complete adjoint campaign therefore remains sealed.

A separate immutable diagnosis shows integrated mid/bottom pressure energy 1.029/0.947 times truth and only 0.0159/0.0193% of model energy in $k \geq 10$, classifying a localized small-amplitude tail rather than a runaway. Job 304125 attempted the 360-day, all-level, three-seed split-1 attribution but is scientifically invalid: each ten-day model call was compared with the next daily snapshot rather than the snapshot ten days later. Its 10–90-day primary RMSE ratios 3.48–4.48 times persistence exposed the temporal mismatch; the output is rejected provenance only. Corrected contract `config/model_c_anomaly_direct_training_spectral_attribution_v2.json`, SHA-256 `29c344310e50...a3de1`, fixes stride 10 and 36 records whose complete t through $t+360$ windows remain in split 1. It reads no validation, inference, response, or adjoint state. Corrected V100 job 304177 completed normally in 1m11s. All three seeds reproduce the tail: day-360 mid/bottom ratios span 12.25–13.54/4.96–5.69, while integrated energy remains 1.02–1.07 times truth and 10–90-day primary ratios remain 0.19–0.37 times persistence. The result authorizes exactly one training-only fine-tune. Contract `config/model_c_anomaly_direct_deep_pressure_spectral_regularization_v1.json`, SHA-256 `4f3377382dc0...acef37`, adds only a 10^{-3}-weighted physical-PHIHYD high-mode error to the unchanged direct-state Model C. It fixes 1,920 low-learning-rate steps, five checkpoints, the same 36 split-1 selection records, strict spectrum/energy/short-skill guards, and no validation or later archive read.

The subsequent spectral fine-tune, LayerNorm/dropout factorial, three-layer/no-padding, and group-specific-head controls all failed their frozen gates. Group-head job 304708 completed normally in 1h30m46s. Diagnostic-best step 14,880 protects primary skill and improves source mid/bottom ratios 13.37/5.57 to 9.21/4.40, with onset delayed to days 220/350 and energy 1.004/0.995, but it still misses factor four. Original seed 20260724/step 13,440 remains selected.

1 Status legend

Status	Meaning
Complete	Finished and checked against the stated acceptance evidence.
Active	Workstream is open and may contain completed substeps.
Next	Immediate executable step on the critical path.
Pending	Planned, but intentionally not started.

2 Milestone dashboard

ID	Milestone	Status	Evidence / exit condition
M0	Reproducible MITgcm build and segmented runner	Complete	S0 build job 283675; four-rank executable and immutable per-segment provenance.
M1	S0 control trajectory	Complete	100-year spin-up plus 10-year daily production; jobs 283676–283686; final data inventory verified.
M2	Tutorial validation	Complete	Validation job 283958; seven of seven gates passed; four project-facing figures and machine-readable report.
M3	S1/S2 forcing regimes	Complete	Low- and high-wind adjustment/production jobs 283959–283962 complete; daily inventories verified.
R0	Git provenance baseline	Complete	Clean first commit pushed to the project remote; generated data, upstream clones, builds, environments, and scheduler logs excluded.
M4	Shared ML dataset	Complete	Jobs 284041/284042 built and independently validated the 5.3 GB S0–S2 Zarr store: $3 \times 3600 \times 46 \times 62 \times 62$, centered/masked state, split hashes, and training-only normalizers.
M4b	Version-2 trajectory coverage	Complete	Jobs 290439/290443/290444 completed all exact extensions, the 11 GB successor store, and independent quality validation. Training-only job 290446 passed the predeclared two-times temperature/SSH effective-coverage target; fresh validation and inference remain sealed.
M5	A0 adapted Bire baseline	Complete	Jobs 284043/284050/284063/284065 completed the overfit, development, frozen model, and outputs. A0 improves held U/V forecasts but persistence remains better for temperature and SSH; metrics and figures are saved.
M6	A–D model ladder	Active	A0/A/B are frozen. Pointwise-anomaly/direct-state seed 20260724/step 13,440 is the replicated median and best deterministic Model C. Job 304736 confirms that it beats persistence and climatology through day 200 under S0, but rejects Bire-style day-2,000 stability because ensemble-wide runaway develops. Retain it for 10–30-day forward/adjoint use while separating long-term statistical stabilization from the learned-physics campaign.
M7	I1/I2 wind interpolation	Pending	Two unseen intermediate-wind trajectories generated and opened for inference only.
M8	Forward-response campaign	Pending	224 ten-day perturbed restarts: 128 training, 32 validation, and 64 inference.
M9	Blind adjoint evaluation	Pending	Frozen artifacts/checksums recorded before 12 true-adjoint tests are generated or inspected.
M10	Control-update verification	Pending	Three forward checks demonstrate whether emulator-gradient controls improve each objective.

3 Completed simulation record

3.1 S0: control wind, $\tau_0 = 0.100\ \mathrm{N\,m^{-2}}$

Item	Recorded result
Execution	Build job 283675; spin-up jobs 283676–283685; production job 283686.
Model duration	100-year spin-up plus 10-year production: 110 model-years, using a 360-day year.
Production inventory	3,600 daily `dynState` records and 3,600 daily `surfState` records.
Measured size	4.6 GB in the raw MITgcm archive.
Validation	Job 283958; all seven tutorial gates passed.
Surface heat flux	Mean $-1.9589\ \mathrm{W\,m^{-2}}$.
Potential temperature	Final representative means: 17.6918°C at the surface, 8.3589°C at 305 m, and 2.4327°C in the abyss.
Gyre transport	First production-year extrema: +32.679 and −30.497 Sv.
Sea-surface height	First production-year range: −0.987 to +0.762 m.

The temperature structure, double-gyre transport, sea-surface-height pattern, and late-spin-up trends are consistent with the intended official tutorial behaviour. The deep ocean remains the slowest-adjusting component, as expected; this is recorded as a scientific caveat rather than a failed gate.

3.2 S1 and S2: off-control wind regimes

Run	τ_0	Jobs	Schedule	Daily output	Size
S1	$0.075\ \mathrm{N\,m^{-2}}$	283959 / 283960	5-y adjust + 10-y production	3,600 `dynState` + 3,600 `surfState`	2.7 GB
S2	$0.125\ \mathrm{N\,m^{-2}}$	283961 / 283962	5-y adjust + 10-y production	3,600 `dynState` + 3,600 `surfState`	2.7 GB

Both branches restart from the validated S0 year-100 state. Their five-year adjustment periods are kept separate from the ten-year daily production records used by the ML pipeline.

3.3 S0–S2 exact trajectory-v2 continuations

Run	Array task	Elapsed	Iteration join		Recorded result
S0	290439_0	15m31s	2,851,128 2,851,200	to	3,600 daily field pairs, final pickup, normal end
S1	290439_1	15m28s	2,980,728 2,980,800	to	3,600 daily field pairs, final pickup, normal end
S2	290439_2	15m10s	2,980,728 2,980,800	to	3,600 daily field pairs, final pickup, normal end

The exact joins are one model day (72 iterations) after each immutable v1 endpoint. Every continuation contains 3,600 `dynState` and 3,600 `surfState` metadata/data pairs with no missing daily iteration. Logs end in `STOP NORMAL END`; the floating-point status flags printed at shutdown match the previously accepted production runs and accompany no NaN, fatal, or model error. The three new pickup data SHA-256 values begin `c949...315fb`, `06be...8087`, and `e5a8...1744`.

3.4 Retired high-resolution Bire reconstruction

The earlier 0.25° Bire-oriented spin-up, formerly Slurm job **283514**, was cancelled by the project owner on 21 July 2026. Its scratch tree, compiled build, MITgcm input deck, scheduler wrappers, conversion/training wrappers, and generated reports were subsequently deleted. This campaign is closed and is not a prerequisite for A0 or AF–FNO. Only recovered source and documentation that provide architectural evidence for A0 are retained.

4 Data locations and provenance

Asset	Canonical location
S0 project entry point	`outputs/af_fno/S0/`
S0 result catalog	`outputs/af_fno/S0/results_catalog.json`
S0 validation	`outputs/af_fno/S0/tutorial_validation/`
S0 provenance copies	`outputs/af_fno/S0/provenance/`
S0 raw archive	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S0/`
S1 raw archive	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S1/`
S2 raw archive	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S2/`
Shared ML dataset	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.zarr`
Dataset manifest	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.manifest.json`
Dataset quality report	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.quality.json`
Trajectory-v2 raw continuations	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm_v2/`
Trajectory-v2 dataset	`/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2.zarr`; metadata SHA-256 `ae7be47c4569...78b8cc`

Asset	Canonical location
Trajectory-v2 quality gate	/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2.quality.json; SHA-256 a6f7e5234ace...cf752
Trajectory-v2 effective coverage	/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2_coverage_v1/; report SHA-256 7758cfb64e73...18e5
Pressure-postprocessor validation	outputs/af_fno/pressure_validation_v1/pressure_validation.json
Streamfunction consistency audit	V1 collocation-warning record under outputs/af_fno/streamfunction_consistency_v1/; corrected collocated v2 contract config/streamfunction_consistency_v2.json, SHA-256 88ff2f20fe1a...dea170; report/arrays SHA-256 14fa2bfb2d2b...eae9a7/f807d3025186...c55773; figures and manifest-content SHA-256 d62eed150a13...e6162 under outputs/af_fno/streamfunction_consistency_v2/
A0 overfit artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/overfit_v1/
A0 development artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/development_v1/
A0 frozen artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/final_v1/
A0 figures / numerical outputs	outputs/af_fno/A0/final_v1/
A0 paper-style output package (job 284150 complete)	outputs/af_fno/A0/paper_style_v1/
Model A overfit artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/overfit_v1/
Model A development artifacts (job 284152)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/development_v1/
Model A frozen artifacts (job 284153)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/final_v1/
Model A figures / numerical outputs (job 284155 complete)	outputs/af_fno/A/final_v1/
Model B overfit artifacts (job 284302 complete)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/overfit_v1/
Model B development profiles (jobs 284304–284306 complete)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/development_*_v1/
Model B frozen artifacts (job 284342 complete)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/final_v1/
Model B figures / numerical outputs	outputs/af_fno/B/final_v1/
Model C training-only diagnostics (jobs 284850/284857)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/training_diagnostics_v1/ and /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/training_diagnostics_v2/

Asset	Canonical location
Model C GPU loss-gradient calibration (job 284858)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/loss_calibration_v1/
Model C CPU loss-gradient replica (jobs 284860/284864/285116)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/loss_calibration_cpu_v1/
Model C frozen loss manifest	config/model_c_loss_v1.json
Model C memorization/reload gate (job 285192)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_v1/
Model C controlled duration gate (job 285265)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_duration_v2/
Model C width-64 GPU capacity gate (job 285325)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_width64_gpu_v1/
Model C late-checkpoint audits (job 287581)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/late_checkpoint_objective_audit_width32_v1/ and /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/late_checkpoint_objective_audit_width64_v1/
Model C accepted optimization manifest	config/model_c_optimization_v1.json
Model C accepted memorization gate (job 287583 task 0)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_v1_lr_decay_v1/
Model C rejected loss-v2 manifest / gate (job 287583 task 1)	config/model_c_loss_v2.json and /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_increment_loss_v2/
Model C CPU fallback scaling (job 287556)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/benchmarks/C/cpu_scaling_width64_v2/
Model C validation-search contract	config/model_c_validation_search_v1.json; SHA-256 9e1d44299ae6...c4c6f6
Model C validation-search stages	Jobs 287604, 288466, 289379, and 289915; immutable root /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/validation_search_v1/; stage-manifest SHA-256 values 40b54530d7d9...12d2f05, a2d639f8940c...d5e382, 9bbd4fc4ce3c...f30e17, and 453c05ec408f...2fea2
Model C rejected three-seed gate (job 290382)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/validation_search_v1/final_seed_gate/; decision SHA-256 e19d502442fc...72155, freeze-contract SHA-256 a44622cee685...73628
Model C data-adequacy contract / job 290415	config/model_c_data_adequacy_v1.json; SHA-256 e185301947b4...53c84a9; report /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/data_adequacy_v1/model_c_data_adequacy_report.json, SHA-256 be672face8c4...4e307ee

Asset	Canonical location
Trajectory-v2 expansion contract	`config/trajectories_v2_expansion.json`, SHA-256 `7ee2ae10330a...fe045ed`; three exact ten-year extensions under the new `mitgcm_v2/` scratch root
Trajectory-v2 coverage contract / job 290446	`config/trajectories_v2_coverage_audit.json`, SHA-256 `e539ca1bb61a...9197f36`; immutable report root `/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2_coverage_v1/`
Model C successor-training contract / jobs 290448/290597	`config/model_c_successor_training_v1.json`, SHA-256 `3e0a300e73ec...159285`; immutable checkpoints/reports under `/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/successor_training_v1/`; lightweight results under `outputs/af_fno/C/successor_training_v1/`
Model C successor validation / jobs 290673/290738	Corrected `config/model_c_successor_validation_v2.json`, SHA-256 `56bb6ec59799...0cf27b`; complete rejected report SHA-256 `9853ea7aff21...663fb`; immutable scratch root `/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/successor_validation_v1/`; lightweight training and validation evidence under `outputs/af_fno/C/successor_validation_v1/`
Model C validation figures	Five hash-bound PNGs and `figure_manifest.json` under `outputs/af_fno/C/successor_validation_v1/fresh_validation/`; manifest content SHA-256 `8b24b6af0686...a0845f`
Model C rollout diagnosis / job 291102	`config/model_c_rollout_diagnosis_v1.json`, SHA-256 `d0ecee0db2b2...ff7ba`; complete report/arrays SHA-256 `7efdb0726125...171374/c3c403ab81b2...5f255`; immutable scratch root `/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/rollout_diagnosis_v1/`; three figures and lightweight summary under `outputs/af_fno/C/rollout_diagnosis_v1/`
Model C exact-replay checkpoint audit / job 291164	Completed in 1h19m14s; exact step-14,880 replay, no passing checkpoint, diagnostic best step 14,400. Contract SHA-256 `494ce9de6688...06682`; report/arrays SHA-256 `7fd670f482c5...940367/7983cb5e0c74...ee9227`; published root `outputs/af_fno/C/checkpoint_replay_audit_v1/`
Model C pushforward objective / job 291612	Completed scientific rejection in 20m37s; exact selected step 1,920, worst primary AUC/slow-lead ratios 1.692/3.994. Contract SHA-256 `406ee5248fb4...bccc2`; report/arrays SHA-256 `631cec8e5f49...ebd07/a4469e3c2d3d...b0855`; published root `outputs/af_fno/C/pushforward_objective_v1/`
Model C pushforward duration / job 291616	Completed in 51m31s with bitwise pushforward-v1 replay and exact reload; all eight checkpoints fail, and diagnostic-best step 5,760 has worst primary AUC/slow-lead ratios 1.597/3.609. Contract SHA-256 `4867fa77d68e...78c79`; report/arrays SHA-256 `86bf00b800d6...e9d7/46129f2653b7...7bfc`; published root `outputs/af_fno/C/pushforward_duration_v1/`

Asset	Canonical location
Model C truncated-unroll objective / jobs 291769/291778	Job 291769 stopped before training on a checkpoint-field schema mismatch. Corrected job 291778 completed in 20m01s but failed at diagnostic-best worst primary AUC/slow-lead ratios 1.933/3.701. Contract SHA-256 `0c5aabadbe51...92e3c`; report/arrays SHA-256 `86f7981e7a0a...b5b830/0c5222cd6938...d716`; figure/CSV/summary under `outputs/af_fno/C/truncated_unroll_objective_v2/`, manifest-content SHA-256 `cccd8e757495...ae4c03`
Model C slow-field bias/projection audit / job 292064	Completed in 7m09s; feedback-amplification classification, fixed-bias SST/SSH explained energy 33.43%/27.21%, and 19.25% combined worst-AUC reduction. Contract SHA-256 `fd7d39f5c6a4...05428`; report/arrays SHA-256 `e66b263c826c...0f90/4e7f73c2384e...8547`; plots/CSV/summary under `outputs/af_fno/C/slow_field_bias_projection_v1/`, manifest SHA-256 `0a2b4fbc8411...c4d`
Model C rollout-conditioned loss v3 / job 292293	Completed in 27m46s; diagnostic step 960, all ten-day groups pass, strict long gate fails at worst primary AUC/slow-lead ratios 1.462/3.634. Contract SHA-256 `37f1157851ac...5a08`; report/arrays/checkpoint SHA-256 `e9af34326c36...7ba11/522fe6145735...edff/e9f962878cb5...970b`; two plots, CSV, summary, and manifest `c323a18285f1...0a6d` under `outputs/af_fno/C/rollout_conditioned_loss_v3/`
Model C pointwise-anomaly direct-state gate / job 303447	Completed in 1h29m32s; selected step 14,400, training worst primary AUC/slow-lead ratios 0.394/0.519, and bounded day-200 speed/SST/surface-P/ρ RMSE 0.00212/0.0288/0.0552. Contract SHA-256 `fa38b41578f5...8acea`; report/arrays/checkpoint SHA-256 `2d840fb91682...feb6d5/d012c2851246...5270/1d4b7cae8249...362e`; six plots/CSV/summary under `outputs/af_fno/C/anomaly_direct_v1/`, manifest-content SHA-256 `4137cc82b350...dbb95`
Model C anomaly-direct seed replication / array 303640	Both tasks completed normally in 1h29m30s/1h26m32s. All three seeds pass; fixed median is seed 20260724, step 13,440, checkpoint SHA-256 `28b4809089a2...1a23`. Frozen contract SHA-256 `dd7e0f92ef50...d8325f`; summary/CSV/plot/manifest under `outputs/af_fno/C/anomaly_direct_replication_v1/`
Model C anomaly-direct held-inference forward gate / job 303993	Completed normally in 3m00s. Day-10/30 readiness and primary 10–90-day AUC pass; one-year state is finite/bounded and wind slope passes. Combined gate fails only at mid/bottom-P/ρ spectral ratios 22.29/9.63. Contract SHA-256 `d11029fce3aa...c2f6c`; complete metrics, arrays, and plots under `outputs/af_fno/C/anomaly_direct_forward_v1/`
Model C deep-pressure spectral diagnosis	Frozen failure retained. Integrated mid/bottom day-360 energy is 1.029/0.947 times truth, while $k \geq 10$ contributes only 0.0159/0.0193% of model energy. JSON/CSV/plot/manifest under `outputs/af_fno/C/anomaly_direct_forward_spectral_diagnosis_v1/`; diagnosis-content SHA-256 `d44da39696a3...7a05`
Model C training-only spectral attribution v1 / job 304125	Scheduler-complete in 1m07s but scientifically invalid: truth used stride one daily snapshot per ten-day call. Results under `outputs/af_fno/C/anomaly_direct_training_spectral_attribution_v1/` are rejected provenance only

Asset	**Canonical location**
Model C corrected training-only spectral attribution v2 / job 304177	Completed normally in 1m11s. All seeds reproduce the tail: day-360 mid/bottom ratios 12.25–13.54/4.96–5.69 with integrated energy 1.02–1.07 times truth and primary 10–90-day ratios 0.19–0.37 times persistence. Report/manifest content SHA-256 `f3cb1bd6de11...141927/952eb9ee49f4...243b8`; JSON/NPZ/two plots under `outputs/af_fno/C/anomaly_direct_training_spectral_attribution_v2/`
Model C deep-pressure spectral regularization v1 / job 304324	Completed normally in 10m25s; no checkpoint passes and source step 13,440 remains selected. Step 1,920 improves day-360 mid/bottom ratios 13.37/5.57 to 6.19/2.97, removes the bottom failure, delays mid onset to day 280, keeps energy 1.006/0.980 times truth, and protects primary skill. Report/manifest content SHA-256 `8065faafeecf...ddb75/280a863a5892...f3a8`; JSON/NPZ/two plots under `outputs/af_fno/C/anomaly_direct_deep_pressure_spectral_regularization_v1/`
Model C Bire LayerNorm/dropout controls v1 / array 304376	All tasks completed normally in 1h09m29s–1h23m33s; no arm passes. Diagnostic-best LayerNorm/dropout/combined day-360 mid/bottom ratios are 9.74/4.92, 37.59/15.04, and 19.51/8.27, with worst primary degradation 18.6%, 57.4%, and 63.2%. Aggregate contract/report/manifest content SHA-256 `c977c196b487...eaea/9f7b7cbbf5dc...5527/fb6e622fef1f...53b2`; original Model C retained; per-arm and aggregate results/plots under `outputs/af_fno/C/anomaly_direct_bire_regularization_controls_v1/`
Model C Bire three-layer/no-padding control v1 / jobs 304526/304597	Job 304526 completed training but failed before metrics on an evaluator reconstruction mismatch. Zero-retraining recovery job 304597 completed normally in 2m41s; no checkpoint passes. Diagnostic-best step 14,400 gives day-360 mid/bottom ratios 12.51/5.84, onset days 210/310, bounded energy 1.032/1.011, and 13.1% worst primary degradation. Recovery contract SHA-256 `9232cea2ae58...165b`; report/manifest content SHA-256 `a429db0f64bb...58416/152190e6d06f...14539`; results under `outputs/af_fno/C/anomaly_direct_three_layer_no_padding_v1/three_layer_no_padding/`
Model C group-specific-head control v1 / job 304708	Completed normally in 1h30m46s; no checkpoint passes. Diagnostic-best step 14,880 protects primary skill at 5.5% worst source-relative degradation and improves source mid/bottom day-360 ratios 13.37/5.57 to 9.21/4.40, onset days 220/350, and energy 1.004/0.995. Report/manifest content SHA-256 `fa25da048b5d...0f3b6/8e9c6a867727...5488e`; results under `outputs/af_fno/C/anomaly_direct_group_heads_v1/group_specific_heads/`
Model C S0 day-2000 truth continuation v1 / job 304735	Completed normally in 6m59s with scheduler exit 0, 2,160 paired daily records, final year-126 pickup, and normal termination on all ranks. Contract SHA-256 `dc204636660d...d75ea5`; result/manifest SHA-256 `9ebd446ff6f3...99925/e15530c67dbe...41c6f`; immutable evaluation-only root `/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm_long_truth_v1/S0/production/years_120_126/`

Asset	Canonical location
Model C S0 Bire Figure 3–8 suite v1 / job 304736	Completed normally in 1m07s; all six figures and the complete NPZ/CSV/JSON package were generated for 15 fixed S0 inference starts. Day-200 model speed/P/ρ/SST RMSE is 0.00270/0.0151/0.0241 and beats both baselines; day-2,000 RMSE is 0.426/2.593/5.339 and rejects long-term stability. Arrays/report SHA-256 `30abd95810d3...3c7/32ac2e051726...bd18`; manifest-content SHA-256 `593dd43de772...57d1`; package under `outputs/af_fno/C/anomaly_direct_bire_s0_inference_v1/`
Model C S0 boundary/checkpoint diagnosis v1 / job 304740	Completed normally in 1m46s with zero retraining. Every checkpoint crosses mean normalized amplitude 20 by day 880; earlier steps cross on days 410/580/780 and do not cure the runaway. At selected step 13,440 the 24.9%-of-cells boundary band contains 72.8/85.2/72.7% of day-2,000 $Q_x/Q_y/\psi$ squared error, while timing does not consistently lead the interior. Report/summary/arrays SHA-256 `bc62501a18f7...2fc67/c6c069e52ee5...fef50e/d37d570337a3...b9942ae`; five plots and complete evidence under `outputs/af_fno/C/anomaly_direct_bire_s0_boundary_checkpoint_v1/`
Model C S0 stability instrument v1	Completed post-hoc without a rollout, new state read, training, or selection. Days-300–600 speed/P/ρ/SST gain per ten-day call is 1.0332/1.0397/1.0456 and normalized maximum gain is 1.0252. Contract SHA-256 `ab6d597843b1...fc5be7`; report/content/manifest-content SHA-256 `5212f3aee081...72040/945495f0e096...6cfbe/f03aeb9487ab...869a`; three plots, CSV, JSON, and README under `outputs/af_fno/C/anomaly_direct_s0_stability_instrument_v1/`
Model C S0 stability/tangent comparison v1 / job 304750	Operationally failed after 1m42s. Preflight and model rollout began normally, but explosive float32 squared-error/spectral reductions overflowed and the log-fit rejected the resulting non-finite series. No transactional output package was published and no scientific comparison is assigned. Original contract SHA-256 `5d32d472509a...bca4`; failed-log SHA-256 `ae476ab3b0c...1356`
Model C S0 stability/tangent recovery v2 / job 304751	Completed normally in 2m12s. Every state remains finite, but the prior residual map reaches day-2,000 RMSE $3.32 \times 10^{25}/6.76 \times 10^{26}/1.41 \times 10^{27}$ for speed/P/ρ/SST. LayerNorm lowers selected day-2,000 error to 5.1/21.3/7.9% but remains 5.54/6.48/10.12 times climatology and supercritical in every growth window. High-k power remains 25.5/38.9/77.0/31.5 times truth. Contract/arrays/report/summary SHA-256 `8bc5c0937d0e...e1ac/95e0a42bc437...01175/684815b7882b...7b84/bdd7c1a06c0d.` five plots and complete package under `outputs/af_fno/C/anomaly_direct_s0_stability_tangent_recovery_v2/`
Model C S0 high-k damping control v1 / job 304753	Completed normally in 1m07s; no nonzero strength passes and conditional inference stays sealed. At $\alpha = 0.02$, day-1,000 worst RMSE/climatology, normalized amplitude, and high-k power fall to 0.7768/0.7742/0.4448 of source, but worst protected short RMSE is 1.0658 times source. Stronger damping damages short skill further. Contract/arrays/report/summary SHA-256 `cf2f1c179d36...9554/88eac7163328...9d/8ca08f114763...a66/3c100081bd10...80` package under `outputs/af_fno/C/anomaly_direct_s0_highk_damping_control_v1/`

Asset	Canonical location
Model C S0 reflected spectral-gate control v1 / job 304754	Completed normally in 1m06s; no nonzero strength passes and inference remains sealed. At $\alpha = 0.25$, day-1,000 amplitude/high-k power/worst RMSE fall to 0.4365/0.0638/0.6178 of source, but worst short RMSE is 6.49 times source and absolute day-1,000 amplitude/RMSE remain 3.65/4.41 times truth/climatology. Contract/arrays/report/summary SHA-256 `69b0100aae27...b8e7/74827af0d9e6...e164/2b739a495fc8...ec53/0e2f11d7fba3..` package under `outputs/af_fno/C/anomaly_direct_s0_reflected_spectral_gate_control_v1/`
Model C Arm R reduced-channel causal control v1	Frozen and preflight-complete; results pending. Ten Bire-facing outputs replace the 46-channel state while the selected width-128 four-block direct-state architecture, optimizer, checkpoints, and three-step objective form remain fixed. The 2.2-GB derived cache is bitwise reconstruction-checked; logical-state SHA-256 `b24728abf50f...c5aa`. Contract SHA-256 `a1b8ce50d1b...5a6c5`; one GPU run will train/select on split 1 and then generate the same fixed S0 Figure 3–8 suite through day 2,000 under `outputs/af_fno/C/anomaly_direct_reduced_channel_bire_s0_inference_v1/`
Model C Bire-style figures / job 291134	Completed frozen descriptive contract `config/model_c_bire_figures_v1.json`, SHA-256 `791c84f12c2d...eb696b`; report/arrays SHA-256 `b704b921444d...e300ac5/a73ac9bf2cc9...20b1`; two PNGs and hash manifest under `outputs/af_fno/C/bire_figures_v1/`, manifest-content SHA-256 `258be6fabe13...827f`
Model C Figure-4 baseline companion	Frozen `config/model_c_bire_baseline_zoom_v1.json`, SHA-256 `2a2324090b83...b2a4a`; full/zoom SST and surface-P/ρ figure plus complete CSV under `outputs/af_fno/C/bire_baseline_zoom_v1/`, manifest-content SHA-256 `01433936e406...99063`
Model C per-lead streamfunction maps / job 291135	Completed descriptive contract `config/model_c_bire_streamfunction_leads_v1.json`, SHA-256 `ccb965e58bdc...ae1934c`; report/arrays SHA-256 `71c6e1dc1f9a...89c206/b4126494d9c0...3bcd48`; eight PNGs and manifest-content SHA-256 `3b74d759d551...450ac7` under `outputs/af_fno/C/bire_streamfunction_leads_v1/`

The S0 project directory contains a symbolic link named `raw_mitgcm`; bulk output is not duplicated in `/home`. The catalog records counts, final pickup, hashes, executable provenance, and canonical paths. The shared-dataset manifest seals the three source archives, the fixed split, and normalizer hashes; its independent quality report records finite/mask/spectral checks, one-day increment ranges, and a fresh normalization recomputation.

5 Frozen project decisions

- The active scientific benchmark is the official 1°, $62 \times 62 \times 15$ baroclinic double gyre. Exact recovery of the paper's hidden 0.25° numbers is outside the primary claim; qualitative trends are the reproduction target.
- Model calendars use 360 days per year. The time step is 1,200 s, or 72 steps per model day.
- Standard trajectory production uses four MPI ranks. Long runs are split into restart-safe jobs that fit

the two-hour `short` partition.

- S0–S2 share physics, grid, preprocessing, state definition, splits, and metrics. Only the prescribed forcing regime differs.
- `PHIHYD` remains an exact configured hydrostatic reconstruction from all 15 temperature levels and SSH. Streamfunction is labeled the U-derived barotropic transport streamfunction. The collocated raw-face U/V audit passes daily and annual path consistency in S0–S2 by orders of magnitude, but the linear-free-surface instantaneous-divergence caveat remains explicit.
- True MITgcm adjoints are sealed test data. They are excluded from training, early stopping, architecture selection, hyperparameter tuning, perturbation-scale selection, and loss selection.
- All models must use the same immutable train/validation/test partition and training-only normalization statistics so architecture and loss comparisons remain interpretable.
- Model C validation-search v1 is closed as a scientific rejection. Its three-seed decision records `configuration_frozen=false` and `inference_authorized=false`; no later-data archive may be opened on the strength of the near-threshold SSH results.
- Git stores source, configuration, tests, small manifests, and project-facing documentation. Raw/reduced arrays, checkpoints, upstream clones, virtual environments, compiled trees, and scheduler logs remain external to Git.

6 Baseline and forward-selection ladder

Decision added 21 July and revised 23 July 2026. Establish an adapted Bire paper-architecture baseline, A0, before developing the modern NeuralOperator A–D sequence. This tests whether the published architectural idea recovers the paper's qualitative wind and forecast trends on the available tutorial configuration. Keep B frozen, select C only for forward fidelity, and add local-response supervision only in D.

A0 is a controlled adaptation, not a claim of exact historical reproduction. It keeps the recognizable Bire choices—channel-diagonal/separable spectral convolution, explicit pointwise channel mixing, three Fourier blocks, hidden width 128, positional encoding, and one-step state loss. Resolution-equivalent Fourier truncation uses 16 modes on the 62×62 grid. The frozen A0 did not use the paper code's stride-8 coarse pretraining; this is a recorded adaptation, not an inferred reproduction.

Changes forced by this benchmark must be disclosed: 46 Markov-state channels (centered U, V, temperature, and SSH), the 62×62 grid, ten production years per regime, wind stress as an input, and a ten-day primary comparison horizon. The obsolete public repository will be used as architectural evidence, not executed unchanged. Pressure is not a forgotten channel: evaluation contract v2 reconstructs MITgcm `PHIHYD` at levels 0, 7, and 14 from predicted temperature and SSH using the configured linear EOS and exact finite-difference hydrostatic operator. Its archived S0 validation passes with $3.89 \times 10^{-4}\ \mathrm{m^2\,s^{-2}}$ maximum error. The velocity-derived circulation diagnostic is independently verified under `config/streamfunction_consistency_v2.json`: daily U/V path p90 relative RMSE is at most 3.25×10^{-5}, annual absolute mismatch is 4.74–6.00×10^{-5} Sv, correlation is effectively one, and all boundary normal transports are zero. This supports the U-derived transport streamfunction strongly while retaining the free-surface caveat.

The comparison ladder is therefore:

Persistence / climatology $\longrightarrow$ A0: adapted Bire $\longrightarrow$ A: modern state-only $\longrightarrow$ B: forward losses $\longrightarrow$ C: forward-selected $\longrightarrow$ D: AF–FNO response supervision.

Every transition changes a declared element, making gains attributable rather than confounding the dataset, architecture, and derivative-aware objective at once.

7 Immediate execution sequence

Current position: boundary mechanism classified; replace the short-horizon boundedness instrument before another model change. Zero-retraining job 304740 completed normally and verified that all seven seed-20260724 checkpoints develop the long-run runaway. Earlier checkpoints cross mean maximum normalized amplitude 20 on days 410/580/780, while the selected step 13,440 crosses latest at day 880; training duration is not the cure. At day 2,000 the four-cell boundary band contains 72.8/85.2/72.7% of $Q_x/Q_y/\psi$ squared error despite containing only 24.9% of wet cells. The pattern is reproducible across all 15 starts, but boundary growth does not consistently precede interior growth. The classification is therefore a systematic boundary-amplified global transport mode, not a single initialization or a proven boundary-to-interior cascade. The complete hash-verified package and five plots are under `outputs/af_fno/C/anomaly_direct_bire_s0_boundary_checkpoint_v1/`. The post-hoc stability instrument is now complete. Over days 300–600, speed/P/ρ/SST RMSE gains per ten-day call are 1.0332/1.0397/1.0456 with bootstrap intervals wholly above one; normalized maximum amplitude gains 1.0252 per call. Replace finiteness by saturation and statistical-climate gates before another candidate. The next bounded GPU comparison is now frozen and preflight-complete under contract `5d32d472509a...bca4`. It applies the corrected instrument to the retained source, the already-trained LayerNorm diagnostic, and the prior residual map for the same 15 S0 members and 200 calls. Shared-coordinate singular and ten-call tangent gains are computed only for the two direct maps and are not called spectral radii. No retraining, selection, or promotion is permitted. Job 304750 then exposed a reduction-level failure: one comparator became large enough to overflow float32 physical diagnostics, after which the log-fit rejected the non-finite curve. It published no package. Recovery v2 is now frozen under `8bc5c0937d0e...e1ac`; it retains the entire model/rollout/tangent protocol, uses float64 only for physical reductions, records memberwise first non-finite state lead, and censors incomplete windows. Results remain pending and the recovery still cannot select or promote a checkpoint. Recovery job 304751 completed normally in 2m12s and changes the decision. All maps stay finite, so finiteness is rejected as a gate. The prior residual comparator caused the float32 overflow and grows by roughly 1.38–1.42 per call; it is not contracting. LayerNorm reduces selected day-2,000 speed/P/ρ/SST error to 5.1/21.3/7.9%, but still lies 5.54/6.48/10.12 times climatology and every fitted window remains supercritical. Tangent gains show specific high-k damping but larger full/mid ten-call gain, rejecting a uniform-contraction explanation. Retain selected Model C for short work, promote no comparator, and test a bounded high-k/gated-normalization mechanism next. That next causal control is now frozen under `cf2f1c179d36...9554`. It trains no weight: weak reflected binomial damping strengths are selected solely on ten split-1 S0 day-1,000 rollouts with a 5% short-skill guard and strict amplitude, RMSE, and high-k improvement requirements. The fixed day-2,000 S0 archive opens only after a nonzero strength passes every training gate. Job 304753 completed normally in 1m07s and no strength passes. The weakest $\alpha = 0.02$ intervention meets every day-1,000 reduction gate: worst RMSE/climatology, normalized amplitude, and worst high-k power are 0.7768/0.7742/0.4448 of source. It misses the protected short gate, however, with day-90 SST and speed at 1.0658 and 1.0612 times source. The conditional inference phase therefore remains sealed. This supports high-k content as a contributor to the growth but rejects a fixed isotropic post-step smoother as the solution. Preserve the 5% guard and move to a training-integrated, state- or scale-gated correction. The immediate bounded scale-gated control is now frozen under `69b0100aae27...b8e7`. It uses an even-reflected spectral basis, leaves $k \leq 8$ unchanged, transitions over $8 < k < 10$, and attenuates only $k \geq 10$. Five strengths are judged by the identical split-1 training gate; only a passing nonzero value can open fixed S0 inference. This is zero retraining and cannot promote a checkpoint. Job 304754 completed normally in 1m06s and rejects every strength. The weakest value removes 93.6% of source high-band power and halves normalized amplitude, yet makes the worst short RMSE 6.49 times source and still leaves day-1,000 amplitude/RMSE 3.65/4.41 times truth/climatology. Thus the tail is a coupled symptom/amplifier rather than a sufficient explanation. Close fixed filtering, keep inference sealed, and

1. Preserve the completed stage manifests, candidate reports/checkpoints, final-seed artifacts, and rejection decision without mutation. Treat all GPU tasks as operational successes and the gate outcome as a scientific rejection.
2. Preserve the completed data-adequacy report. Its four checks pass: aggregate training fit, strictly improving chronology, positive SSH validation–training gaps in all three regimes, and fewer than 20 effective slow-state samples. The S1 training failure prevents interpreting this as a data-only diagnosis.
3. Preserve completed jobs 290439/290443/290444/290446 and their immutable raw, dataset-quality, and effective-coverage evidence. The slow-field two-times target passes; do not use the less stable velocity integral-time ratios as a tuning target.
4. Preserve completed training-only successor jobs 290448/290597 under contract SHA-256 `3e0a300e73ec...159285`. The unchanged width-64 data control and isolated $4C$ channel-mixing test both failed only S1 SSH; $4C$ reduced the full-training ratio from 1.103435 to 1.003573 and improved every aggregate group. Width 128 passes every regime/group at worst ratio 0.762683 and every reload/stability check.
5. Preserve completed jobs 290673/290738 and the exact fresh-validation rejection. Surface speed passes every baseline; SST and surface pressure have good day-10 skill but fail persistence over the 10–90-day RMSE curve. Do not reinterpret the amplitude-only 180-day training check as trajectory skill.
6. Preserve completed training-only rollout-diagnosis job 291102 under contract SHA-256 `d0ecee0db2b2...ff7ba`. Every seed has good day-10 slow-field skill but failed 10–90-day AUC and day-90 skill on training chronology, so apply the predeclared objective/gate-mismatch classification. Do not generate more data or open inference.
7. Preserve exact-replay checkpoint-audit job 291164 under contract SHA-256 `494ce9de6688...06682`. Step 14,880 and its six-row history reproduce exactly, but no original checkpoint passes; step 14,400 is diagnostic best. Apply `objective_correction_required`, not a checkpoint-only correction.
8. Preserve rejected pushforward-objective job 291612 under contract SHA-256 `406ee5248fb4...bccc2`. Step 1,920 improves every displayed curve and all source priorities but still fails both slow-field baselines at long lead. Do not replicate or validate it.
9. Preserve completed duration job 291616 under contract SHA-256 `4867fa77d68e...78c79`. Require bitwise replay of pushforward step 1,920 passed, but none of eight extension checkpoints passes. Retain diagnostic-best step 5,760 and the published SST/P/ρ curves; classify terminal-only supervision as structurally insufficient and do not validate it.
10. Preserve completed corrected truncated-unroll contract SHA-256 `0c5aabadbe51...92e3c`. Starting from duration step 5,760, job 291778 supervised three consecutive differentiable calls over the frozen early/late slow-field windows and reloaded nine calls bitwise. It failed every checkpoint and worsened worst primary AUC/slow-lead ratios to 1.933/3.701. Stop before validation and reassess from the saved lead/training curves.
11. Keep Model C's temporal interface explicitly classified as one-input/one-output: 51/46 are channel counts, while each call sees one state and predicts one ten-day residual. Samudra's two-past-state/two-future-state design, which Bire proposed only as future work, is a bounded candidate rather than an immediate replacement. Compare it prospectively against a simpler long-rollout objective/checkpoint correction at matched data and controlled compute.
12. Preserve streamfunction-consistency contract v2 SHA-256 `88ff2f20fe1a...dea170`. It corrects the v1 C-grid face-collocation error without changing the fixed year or thresholds. All S0–S2 daily/annual U/V path, averaging, amplitude, and impermeable-boundary checks pass. Use the U-derived transport-streamfunction label and retain the instantaneous free-surface-divergence caveat.

13. Preserve Bire-style figure contract SHA-256 `791c84f12c2d...eb696b`. It fixes a descriptive 1° streamfunction comparison and 15-member, 0–200-day $\Delta t = 10$ RMSE curves on S2 fresh validation. Persistence repeats the initial condition; climatology is a pointwise split-1 training mean. The figures may characterize the rejected model but may not tune or unseal inference. Job 291134 completed: streamfunction is excellent through day 40, whereas SST and surface pressure leave their requested axes at days 100 and 160 as finite long-rollout drift grows. Preserve both the fixed-axis plots and off-scale numerical archive. Preserve the separate `outputs/af_fno/C/bire_baseline_zoom_v1/` addendum: it changes no source result and makes the SST/P/ρ baseline crossings visible, with a complete CSV for every plotted lead.
14. Preserve per-lead streamfunction-map contract SHA-256 `ccb965e58bdc...ae1934c`. It fixes eight separate maps at days 20–90 for the existing S2 start 6335. State panels share one scale, while each difference panel has a disclosed lead-specific scale plus numerical RMSE/max error. It is descriptive only; do not use the maps to select a checkpoint or unseal inference. Job 291135 completed with all predictions finite: relative streamfunction RMSE rises only from 0.700% at day 20 to 2.211% at day 90, showing that barotropic circulation remains much better preserved than SST and surface pressure.
15. Keep inference, I1/I2, response, and adjoint data sealed. Open them only if an explicitly versioned Model C successor passes a prospectively declared forward gate; train D only from such an accepted design.

8 Remaining simulation and evaluation programme

Work package	Count	Role
S0–S2 trajectory-v2 extensions	3 complete	Exact continuations, conversion, independent quality control, and effective-coverage audit completed in jobs 290439/290443/290444/290446.
I1/I2 trajectories	2	Intermediate wind amplitudes; inference-only interpolation test.
Perturbed forward responses	224	Ten-day local responses: 128 training, 32 validation, 64 inference.
Mandatory true adjoints	12	Three smooth objectives × two horizons (10/30 days) × two unseen states.
Forward control checks	3	Test one emulator-gradient control update for each objective.
Optional true adjoints	6	Ninety-day extension only if mandatory criteria are met.

The expanded programme contains 247 scientific integrations in total: the original five trajectory runs, three authorized S0–S2 extensions, 224 perturbed forward responses, 12 mandatory adjoints, and three forward control checks. The extensions add 30 model-years, taking trajectory accounting from 142 to 172 model-years.

The mandatory adjoints alone are the ground-truth gradient dataset. They evaluate: an interior Gaussian SSH average, a smooth western-boundary SSH average, and an area-weighted barotropic gyre-strength objective. The emulator, evaluation code, pass criteria, and cryptographic hashes must be frozen before these results are opened.

9 Risks, corrections, and controls

Risk / issue	Control or recorded correction
Scratch is not archival	Raw S0–S2 data live on `/bigscratch`. Copy them to durable project storage before the cluster retention deadline; retain catalogs, hashes, manifests, and figures under the project tree.
Trajectory-year arithmetic	The original durations sum to **142**, not 132, model-years: $110 + 15 + 15 + 1 + 1 = 142$. The authorized three ten-year extensions raise the active compute/storage accounting to 172 model-years.
Historical-code incompleteness	Label A0 “adapted paper architecture” and record all forced changes; do not claim a bitwise or numerical reproduction of Bire et al.
Data leakage	Freeze chronological indices and derive normalization only from training years. Keep I1/I2 inference-only.
Adjoint leakage	Enforce the sealed-adjoint contract and checksum frozen model and evaluation artifacts before generating or viewing adjoint results.
Deep-ocean drift	Report depth-dependent trend diagnostics with forecast results; avoid implying full abyssal equilibrium from upper-ocean stationarity.
Large job count	Pack short response integrations into auditable Slurm arrays or bundles while preserving one manifest/result record per scientific integration.
Cross-HPC portability	UCSB absolute paths, modules, accounts, and partitions are not NASA-portable. Introduce site profiles, rebuild dependencies and executables on the destination system, transfer data separately, and pass smoke/restart tests before production.

10 How to update this tracker

Update this document after a milestone changes state, a scientific decision changes, or a material dataset/model artifact is created. For each update:

1. advance the status date and change only milestones supported by durable evidence;
2. record Slurm job IDs, model-time duration, output counts, storage location, and validation result for every completed simulation;
3. add artifact paths and checksums where available; never treat a scratch pathname alone as an archival record;
4. record decision changes in both the relevant section and the change log;
5. regenerate `Project_tracking.pdf` and check the LaTeX log for undefined references, overfull boxes, or missing files.

Operational details and evolving status belong here. Scientific definitions, acceptance criteria, and the intended experimental design belong in `AF_FNO_Project_Plan.tex`; keep the two documents consistent without duplicating extensive derivations.

11 Change log

Date	Type	Change
2026-07-21	Tracker created	Consolidated S0–S2 execution evidence, validation metrics, data locations, active decisions, remaining simulations, and risks.
2026-07-21	Method decision	Added A0, an adapted Bire paper-architecture baseline, before models A–C.
2026-07-21	Accounting correction	Corrected total trajectory duration from 132 to 142 model-years.
2026-07-21	Infrastructure	Prepared the source tree for Git: removed disposable Slurm/LaTeX/Python artifacts, strengthened ignore rules, replaced the paper-only README with the AF–FNO project overview, and documented NASA portability. The first remote commit remains R0.
2026-07-21	Campaign retired	Recorded cancellation of job 283514 and removed the obsolete 0.25° MITgcm scratch data, build, input deck, job wrappers, and reports. Retained only Bire-derived code evidence required for A0.
2026-07-21	Dataset conversion	Added and locally tested the S0–S2 MDS-to-Zarr converter, including declared C-grid centering, wet-mask handling, chronological split codes, and training-only normalizers. All 39 available tests passed (one optional ML test skipped); Slurm job 284041 was submitted to create and validate `trajectories_v1.zarr`.
2026-07-21	M4 complete	Conversion job 284041 completed in 1m32s and produced the 5.3 GB $3 \times 3600 \times 46 \times 62 \times 62$ store. Independent job 284042 completed in 48s; it verified finite values, masks, split contract, spectrum, daily continuity, and recomputed the training-only normalizers to 3.12×10^{-7} mean and 1.56×10^{-7} scale maximum absolute error.
2026-07-21	A0 gate started	Added and smoke-checked the adapted A0 loader/trainer: 46 normalized state inputs, one normalized wind input, 46 normalized next-state targets, the recovered three-block separable FNO with 16 modes, masked paper-style MSE+MAE loss, and bitwise save/reload acceptance. Submitted the 96-sample GPU overfit gate as job 284043.
2026-07-21	A0 overfit passed	Job 284043 completed in 6m11s on one V100 GPU. Its 96 regime-balanced, training-only ten-day pairs reduced the masked normalized loss from 1.04999 to 0.002128 at learning rate 0.01; no fallback was needed, and saved/reloaded inference was bitwise identical. Checkpoint SHA-256: `30406885a1d4...295c83b0e9`.

Date	Type	Change
2026-07-21	A0 development started	Added the sealed-split development runner with chunk-aware chronological batches and persistence comparison. Submitted V100 job 284050 for 12 epochs over 7,530 training pairs and 780 validation pairs; this is a development gate, not the frozen final A0 model.
2026-07-21	A0 schedule frozen	Development job 284050 completed in 18m12s. Its best validation loss was 0.001841 versus persistence 0.014453, with bitwise checkpoint reload; the minimum occurred at epoch 10. Froze the final configuration (10 epochs, batch 8, learning rate 0.01, unchanged paper-style optimizer/loss/seed) and submitted its single final realization as V100 job 284063.
2026-07-21	A0 model frozen	Final job 284063 completed in 14m46s. It preserved bitwise checkpoint reload and obtained losses 0.001841/0.002272 on held validation/inference versus persistence 0.014453/0.014357. Checkpoint SHA-256: `b879b64b8cc7...3663be18f`. Submitted job 284065 to save the numerical and figure outputs in the project-facing output tree.
2026-07-21	A0 baseline closed	Evaluation job 284065 completed in 1m09s and saved `a0_metrics.json`, `a0_metrics_arrays.npz`, and three PNG figures under `outputs/af_fno/A0/final_v1/`. Held one-step A0/persistence RMSE ratios are 0.395 (U), 0.912 (V), 4.771 (temperature), and 13.612 (SSH); 10–100-day rollout figures show the same mixed state-variable result. M5 is complete without a claim that A0 beats persistence for every component.
2026-07-21	A0 paper-style evaluation complete	V100 job 284150 completed in 53s without retraining or changing A0 and saved a distinct output package. Its 15-member-per-regime, 10–200-day diagnostics adapt the forms of Bire et al. Figures 3, 5, 6, 7, and 9 to the 1-degree tutorial: multi-lead snapshots, spatial RMSE, ACC with training-global anomaly reference, and low/control/high wind-regime skill. This is explicitly a diagnostic-form comparison, not a numerical reproduction of the paper figures.
2026-07-21	Model A overfit submitted	After 48 passing project tests (one intentional dependency-path skip), style/diff checks, a 62-by-62 CPU forward/backward pass, and a production-Zarr sample read, submitted V100 job 284068. It ran the declared 96 regime-balanced, training-only pairs for the residual-learning/reload acceptance gate.
2026-07-21	Model A overfit passed	Job 284068 trained for 5m49s but failed only at checkpoint reload because NeuralOperator 2.0 adds a non-parameter `_metadata` entry that PyTorch rejects as a state-dict key. Filtered that entry from portable checkpoints, added a reload unit test, set `CUBLAS_WORKSPACE_CONFIG` for CUDA determinism, and reran the unchanged gate as job 284151. Job 284151 completed in 5m58s: the masked relative-L_2 loss fell from 0.52061 to 0.01589 (best), and save/reload was bitwise exact. Checkpoint SHA-256: `1a8ae303b797...43c21aeaad5b2`.

Date	Type	Change
2026-07-21	Model A development submitted	After 52 passing project tests (one intentional dependency-path skip), submitted V100 job 284152. It trains only the 7,530 chronological train pairs and scores the 780 held validation pairs for 12 epochs, against persistence, to select a single final Model A training duration.
2026-07-21	Model A schedule frozen	Development job 284152 completed in 17m24s. Its best held validation relative-L_2 loss was 0.018119 versus persistence 0.110743 at epoch 10; final epoch loss was 0.019212, and reload was bitwise exact. Froze one final schedule (10 epochs, batch 8, learning rate 0.001, AdamW $\beta = (0.9, 0.95)$, 10^{-5} weight decay, seed 20260721) and submitted V100 job 284153.
2026-07-21	Model A model frozen	Final job 284153 completed in 14m38s. It preserved bitwise checkpoint reload and achieved held validation/inference relative-L_2 losses 0.018119/0.018400 versus persistence 0.110743/0.110719. Submitted job 284155 to save Model A one-step, rollout, snapshot, and direct A0-versus-A figures plus numerical outputs in the project-facing output tree.
2026-07-21	Model A evaluation complete	Job 284155 completed in 1m38s and saved metrics/arrays plus four figures under `outputs/af_fno/A/final_v1/`. Held 10-day Model A/persistence RMSE ratios are 0.169 (U), 0.467 (V), 2.052 (temperature), and 2.277 (SSH), compared with A0's 0.395, 0.912, 4.771, and 13.612. At 100 days Model A still has temperature/SSH ratios 17.79/24.48; therefore A is a forward-state improvement over A0 but not a stable long-rollout solution.
2026-07-22	Model B gate submitted	Implemented the controlled Model B forward-loss pipeline without changing Model A's architecture, data, residual target, optimizer, or seed. The loss contract adds 20/30-day autoregressive error, 12-bin radial anomaly spectra at 10/20/30 days, and the first four wet cells from the western wall at all three leads. Added incremental development profiles, frozen-protocol A0/A/B/persistence evaluation, loss-contract hashing, checkpoint reload gates, tests, and Slurm wrappers. All 62 available tests passed (one intentional skip), and a production-grid three-step CPU forward/backward check had finite gradients. Submitted the 96-sample full-loss V100 acceptance run as job 284302.
2026-07-22	Model B overfit passed	Job 284302 completed in 13m12s. On 96 balanced training-only rollouts it reduced complete loss from 1.52456 to 0.04213 and one-step state loss from 0.52075 to 0.01704; rollout, spectral, and western-boundary components all fell, learning rate 0.001 passed without fallback, and checkpoint reload was bitwise exact. Checkpoint SHA-256: `512707d82bce...17871c159e28`. Dependency-gated rollout-only, rollout-plus-spectral, and full-loss development jobs 284304–284306 started after acceptance.

Date	Type	Change
2026-07-22	Model B schedule frozen	Development jobs 284304–284306 completed in 43–44 minutes with bitwise reload. At each profile's held minimum, common full-objective values were 0.044899 (rollout only), 0.044081 (rollout plus spectra), and 0.043064 (full); the full profile also had the lowest state (0.017925), rollout (0.033599), spectral (0.009012), and boundary (0.031558) terms. Its minimum was epoch 12 versus persistence 0.306148. Froze epochs 12, batch 8, learning rate 0.001, unchanged optimizer/seed/loss weights, and submitted final job 284342. Full-development checkpoint SHA-256: `e185477b126c...af23a7a8119d`.
2026-07-23	Model B closed	Final job 284342 completed in 36m19s on one V100, preserved bitwise reload, and saved the frozen checkpoint. Evaluation job 284346 completed the common 10–100-day package. B reduces A's 100-day U/V/temperature/SSH errors to 0.521/0.409/0.450/0.405 of A, but its own persistence ratios remain 0.449/0.997/8.009/9.925. B is retained unchanged as a successful rollout-loss ablation, not accepted as the forward parent of the response model.
2026-07-23	Complete jobs cancelled	Complete one-year A/B characterization jobs 284816 and 284817 were pending with zero runtime and were cancelled. Their frozen checkpoints and existing evidence remain; uniform final packages are deferred until an accepted C configuration freezes so they are generated once for the final comparison.
2026-07-23	Model ladder revised	Inserted forward-selected Model C between B and the response-supervised model, now named D. The Duruisseaux practical FNO workflow and Bire evidence pattern remain the references. C uses a bounded dense-FNO search and must pass the complete forward gate before response or adjoint work opens.
2026-07-23	Model C C0 started	Added the bounded C architecture/loss contract and training-only diagnostics. Equal U/V/temperature/SSH state and boundary terms, per-channel ten-day increment RMS, and Hann-tapered increment spectra directly address B's aggregate-loss and spectral weaknesses. Eight Model C tests, style checks, shell validation, and a 62 × 62 three-step forward/backward pass succeeded; width-32/64 parameter counts are 630,262/2,443,542. CPU job 284849 exposed an incorrect assertion that checked future pair-start codes rather than target snapshot codes and stopped before creating output. Added a regression test and submitted corrected job 284850 with unchanged train-only scope.

Date	Type	Change
2026-07-23	Model C C0 complete	Corrected job 284850 completed in 42s and verified the immutable dataset hash and train-only read contract. Its 96-sample audit found a 147-fold increment-RMS range and only 278–398 effective increment samples. Because extra meridional modes retained more velocity-increment energy than extra zonal modes, added $(24, 16)$ before validation and reran job 284857 in 33s. At $(16, 16)$, U/V/temperature/SSH increment fractions are 8.8/24.0/67.8/85.6%; at $(24, 16)$, 12.5/32.9/70.2/89.0%.
2026-07-23	Model C C1a submitted	Added a training-only loss-gradient calibration: 20-epoch core warmup on 96 balanced rollouts, followed by raw-term and parameter-gradient audits. It proposes, but does not freeze, auxiliary weights and writes no candidate checkpoint. Nine Model C tests, style, shell, and production-grid checks passed; submitted V100 job 284858.
2026-07-23	CPU C1a replica corrected	CUDA is not scientifically required for the 96-rollout calibration. Job 284860 failed safely after 19s, before output, because its initial audit squared very large gradients in float32. Moving the global norm to float64 prevented overflow, but live job 284864 then revealed that a direct cast discarded imaginary spectral-weight gradients; it was cancelled before output. The final helper preserves real gradients in float64 and complex gradients in complex128 and is covered by a large real/complex regression test.
2026-07-23	Model C C1a complete	Complex-safe CPU job 285116 completed in 4m39s using 1.53 GB maximum RSS. After warmup, state/increment/rollout/spectral/boundary gradient norms were 1.965/830.5/6.484/40,763/7.411. Rejected the clipped 0.01 increment/spectral proposal and froze rounded unclipped coefficients $0.001/0.15/10^{-5}/0.065$, whose weighted gradient ratios to state are 0.423/0.495/0.207/0.245. Report SHA-256: `3406c9104dd6...592e469`. Cancelled redundant pending GPU job 284858 with zero runtime.
2026-07-23	Model C C1b submitted	Added a 96-rollout, training-only memorization gate with fixed initialization/batch order across the bounded learning-rate fallback. Acceptance is groupwise rather than aggregate: 90% total reduction, 50% spectral reduction, maximum state group 0.08, maximum increment group 1.0, rollout 0.10, boundary 0.15, and bitwise-exact three-step reload. Eleven Model C tests and all style/diff/shell checks passed; submitted eight-core short-queue job 285192.

Date	Type	Change
2026-07-23	Model C C1b rejected	Job 285192 completed both 160-epoch attempts in 58m30s and failed only the increment criterion. At learning rate 0.0005, total/spectral reductions were 97.7/99.1%, maximum state group was 0.0304, rollout/boundary were 0.0312/0.0397, and U/V increments were 0.254/0.442 times exact-subset persistence; temperature/SSH remained 1.285/2.032. The absolute-1 threshold is replaced by the mathematically correct per-group ratio to persistence on the same sampled records, without relaxing the requirement.
2026-07-23	Model C C1c submitted	The lower learning rate reached its best total at epoch 160, so undertraining is tested before changing objective or capacity. Added version-2 diagnostics that preserve full learning curves and a reload-tested best checkpoint on rejection. Job 285265 fixes data, architecture, loss, seed, batch order, and learning rate 0.0005, changing only duration from 160 to 320 epochs.
2026-07-23	Pressure evaluation completed	Added a fixed MITGCM-consistent `PHIHYD` postprocessor without changing any model input, output, weight, or split. It integrates the configured linear-EOS density anomaly and adds the linear-free-surface $g\eta$ term. Against the archived S0 `PH` dump at iteration 2,851,200, all-depth wet-cell maximum error is 3.89×10^{-4} and RMSE is 9.75×10^{-6} $\mathrm{m}^2\,\mathrm{s}^{-2}$. Evaluation contract v2 adds surface/mid/bottom pressure curves, maps, snapshots, spectra, boundary metrics, and gate comparisons. Existing v1 packages remain immutable and pre-pressure; frozen inference will be replayed once for the uniform v2 comparison after an accepted C configuration freezes.
2026-07-26	Model C C1c rejected	Job 285265 completed all 320 epochs and every total, spectral, state, rollout, boundary, and exact-reload criterion, but no evaluated epoch passed every increment group. Epoch 315 gave U/V/temperature/SSH persistence ratios 0.206/0.353/0.979/1.219. This is a scientific SSH rejection, although the scheduler recorded exit 1 under the older runner.
2026-07-26	Width-64 capacity rejected	GPU job 285325 completed 320 epochs with exact reload. Its selected epoch ratios were 0.145/0.292/1.083/1.385, and the best balanced epoch still had SSH at 1.302. Four times the parameter count did not solve the slow-field failure, so a further width increase is unsupported.
2026-07-26	CPU fallback measured	Scaling job 287556 completed the same short benchmark at 4, 8, and 16 CPU cores in 436, 272, and 253 seconds. Sixteen cores is the measured fallback if GPU access would materially delay useful work; backend replication remains secondary.

Date	**Type**	**Change**
2026-07-26	Late-objective audit complete	GPU array 287581 completed in 30/33 seconds and independently rechecked all 64 evaluated epochs in each width history, the exact 96 training records, reload provenance, and every memorization criterion. At widths 32/64, weighted increment loss was 5.7/7.2% of state but its gradient norm was already 0.422/0.406 times state; SSH dominated that gradient and was strongly aligned with state. Exact gradient reconstruction errors were approximately 10^{-7}.
2026-07-26	Memorization schedule accepted	Bounded GPU array 287583 isolated two changes. Task 1's versioned 0.0025 increment-weight loss was scientifically rejected, with best balanced SSH ratio 1.328. Task 0 retained loss v1 and applied learning rate 0.0005 through epoch 240 then 0.0001; epoch 315 passed every gate at U/V/temperature/SSH ratios 0.211/0.360/0.794/0.990, 98.24/99.82% total/spectral reductions, 0.0248 maximum state error, 0.0231 rollout, 0.0301 boundary, and exact reload. Only epoch 315 passed every group, so this authorizes bounded validation but does not validate or freeze C.
2026-07-26	Validation contract frozen	Before reading any validation state metric, froze `config/model_c_validation_search_v1.json` with SHA-256 `9e1d44299ae6...c4c6f6`. Ten mode/width candidates undergo four from-scratch rounds with 25/50/75/100% chronology and 1,920/3,840/5,760/7,680 optimizer steps. Ranking first requires all physical U/V/temperature/SSH ten-day ratios below persistence, then minimizes worst-group ratio and a 30/90/180-day full-domain/boundary/spectral score. Conditional padding/layer triggers and final seeds 20260723/20260724/20260725 are fixed; inference and later data remain sealed.
2026-07-26	Validation stages 1–2 complete	GPU array 287604 completed all ten 25%-chronology candidates in 5m58s–8m45s and advanced `m16_24_w64`, `m24_16_w64`, `m24_24_w64`, `m12_12_w64`, and `m12_12_w32`; manifest SHA-256 `40b54530d7d9...12d2f05`. Job 288466 completed the five fresh 50% runs in 10m21s–15m15s and advanced `m16_24_w64`, `m24_16_w64`, and `m12_12_w64`; their worst-group SSH ratios were 1.470, 1.483, and 1.599. Manifest SHA-256: `a2d639f8940c...d5e382`. Every scheduler task and exact-reload check passed; no candidate yet met the scientific gate.

Date	Type	Change
2026-07-26	Validation stages 3–4 complete	Job 289379 completed the three fresh 75% runs in 16m24s–21m51s and advanced `m16_24_w64/m24_16_w64`, whose SSH ratios were 1.153/1.192; manifest SHA-256 `9bbd4fc4ce3c...f30e17`. Job 289915 completed both fresh full-resource runs in 21m18s–28m37s. The $(24, 16)$, width-64 design ranked first at U/V/temperature/SSH ratios 0.1473/0.3547/0.6597/1.00014 and physics score 22.84, ahead of $(16, 24)$, width 64 at 0.1577/0.3723/0.6353/1.02477 and 28.22. Training worst-group ratios 0.915/0.941 and boundary-to-full ratios 0.667/0.639 left both conditional triggers false, so no depth or padding branch opened. Stage-4 manifest SHA-256: `453c05ec408f...2fea2`.
2026-07-26	Three-seed gate rejected	GPU array 290382 trained the selected $(24, 16)$, width-64 architecture afresh for seeds 20260723/20260724/20260725; all tasks completed with exit 0 in 28m39s/22m23s/21m36s and all three checkpoints reloaded bitwise exactly. U/V/temperature beat persistence for every seed, but SSH ratios 1.00014/1.00383/1.03818 failed the strict every-seed rule. The immutable decision therefore records `scientifically_rejected_three_seed_gate`, `configuration_frozen=false`, and `inference_authorized=false`; SHA-256 `e19d502442fc...72155`. Inference, I1/I2, response, and adjoint data remain sealed.
2026-07-27	Freeze data-adequacy gate	Before computing per-regime checkpoint metrics, froze `config/model_c_data_adequacy_v1.json`, SHA-256 `e185301947b4...53c84a9`. It requires aggregate training fit for every seed, a strictly improving four-stage chronology curve, a positive median SSH validation–training gap in at least two regimes, and no more than 20 effective state samples for both temperature and SSH. It reads only split codes 1/2.
2026-07-27	Authorize trajectory v2	GPU job 290415 completed in 3m16s. All four expansion checks passed; state-RMS effective totals are 7.59/7.69 for temperature/SSH, and aggregate SSH block-bootstrap intervals include persistence. S1 is the localized weakness: training ratios 1.32–1.36 and validation ratios 1.50–1.55, versus sub-persistence S0/S2. Report SHA-256 `be672face8c4...4e307ee` authorizes a two-times data expansion but also requires non-data successor changes.
2026-07-27	Freeze expansion design	`config/trajectories_v2_expansion.json`, SHA-256 `7ee2ae10330a...fe045ed`, declares exact ten-year continuations of S0–S2 in a new `mitgcm_v2/` root. The successor store has 7,200 records per regime and 15,060 training pairs; opened v1 validation is excluded, original inference is preserved, and fresh validation/inference blocks retain 90-day buffers. Effective coverage must be measured after conversion.

Date	Type	Change
2026-07-27	Complete v2 simulations	Array 290439 completed all three exact ten-year continuations with exit 0 in 15m10s–15m31s. Each task wrote 3,600 daily state pairs, a final pickup, a gap-free 72-iteration chronology, and `STOP NORMAL END`. No v1 artifact was overwritten.
2026-07-27	Validate trajectory v2	Jobs 290443/290444 built and independently checked the 11 GB $3 \times 7200 \times 46 \times 62 \times 62$ store. The v1 prefix and sampled raw extension records are bitwise exact; finite, land-zero, split, chronology, and train-only-normalizer checks pass. Dataset metadata/quality SHA-256 values are `ae7be47c4569...78b8cc/a6f7e5234ace...cf752`.
2026-07-27	Pass effective-coverage target	Contract SHA-256 `e539ca1bb61a...9197f36` was frozen after quality control and before metrics. Job 290446 found temperature/SSH state multipliers 2.142/2.039 and increment multipliers 2.106/2.253, passing the prospective two-times slow-field target. Report SHA-256: `7758cfb64e73...18e5`.
2026-07-27	Start C successor phase 1	Before v2 training metrics, froze contract SHA-256 `3e0a300e73ec...159285`. Loss, modes, depth, padding, optimizer exposure, and training records remain fixed. GPU array 290448 compares the exact width-64 data-only control with the isolated width-64 $4C$ Channel-MLP ablation; the width-128/256-lift/256-projection candidate is phase 2. Fresh validation remains sealed.
2026-07-27	Complete C successor phase 1	Array tasks 290448_0/290448_1 completed with exit zero in 41m53s/57m49s. Both have exact three-step reload and finite, amplitude-bounded 180-day rollouts. Each fails only S1 SSH: the data-only control is 1.103435 times persistence and isolated $4C$ channel mixing is 1.003573 over all 5,020 S1 training pairs. The $4C$ change improves aggregate U/V/temperature/SSH ratios by 23.6/22.2/9.4/7.9%, so it is beneficial but not sufficient under the exact gate. Reports/checkpoints are hash-addressed in `outputs/af_fno/C/successor_training_v1/phase_1/`.
2026-07-27	Start C successor phase 2	Because both phase-1 reports completed operationally, submitted the already-contracted width-128, lift-256, projection-256, $4C$ candidate as GPU job 290597. It changes no data, loss, modes, depth, padding, optimizer exposure, seed, or training records and does not read fresh validation or any later archive.
2026-07-27	Pass C successor training gate	Job 290597 completed with exit zero in 1h20m22s. Aggregate U/V/temperature/SSH ratios are 0.049040/0.121279/0.384816/0.522413; every group passes separately in S0/S1/S2, with S1 SSH at 0.762683. Exact reload, finite 180-day rollout, and every 90/180-day amplitude check pass. Report/checkpoint SHA-256 values are `5946515ea81a...68433c/2d10e48adc5e...360060`.

Date	**Type**	**Change**
2026-07-27	Freeze successor validation v2	Before any fresh validation state metric, froze seeds 20260723/24/25, all complete 10–90-day starts, Bire-primary surface speed/SST/derived pressure, persistence/climatology/A0 baselines, paired block-bootstrap confidence, and SSH/streamfunction stability. A split-only audit found exactly 180 complete starts per regime, indices 6300–6479; immutable v1 expected 181 and is superseded by v2 SHA-256 `56bb6ec59799...0cf27b` without changing science.
2026-07-27	Start successor seed replication	GPU array 290673 trains only the missing seeds 20260724/20260725 with the accepted width-128 architecture and identical training-only checkpoint-selection rule. Seed 20260723 reuses job-290597's hash-pinned checkpoint. Split 2 and every later archive remain unread.
2026-07-27	Pass three-seed training gate	Array tasks 290673_0/290673_1 completed with exit zero in 1h08m03s/1h09m54s. Seeds 20260723/20260724/20260725 all beat persistence for every U/V/temperature/SSH group separately in S0/S1/S2; their worst ratios are 0.762683/0.732270/0.839507, each for S1 SSH. Every exact-reload, finite-rollout, and 90/180-day amplitude criterion passes. This authorizes one execution of frozen fresh-validation v2, not inference.
2026-07-27	Start frozen fresh validation	Submitted GPU job 290738 to apply the unmodified v2 contract once to all three hash-pinned checkpoints and all 180 complete 10–90-day starts per regime. Split 2 is now authorized for this job only; inference, intermediate winds, response, and adjoint data remain sealed.
2026-07-27	Reject successor fresh validation	Job 290738 completed with exit zero in 13m20s, but no seed passed the full point gate and the paired bootstrap failed. Three-seed mean surface-speed RMSE-AUC is 0.245 times persistence and passes every baseline. SST/surface-pressure ratios are 2.313/2.198: both are good at day 10, cross persistence around days 20–30, and reach about 4.5–5 by day 90. Every seed is finite, land-zero, and passes SSH/streamfunction amplitude and wind-slope checks. The normalized-maximum limit 20 also fails, but validation truth reaches 21.092, so that subgate is non-discriminating; primary failures independently reject C. Report/arrays SHA-256: `9853ea7aff21...663fb/fd84199b585e...dccc`.
2026-07-27	Freeze rollout diagnosis v1	Before reading balanced long-rollout training metrics, froze `config/model_c_rollout_diagnosis_v1.json`, SHA-256 `d0ecee0db2b2...ff7ba`. It evaluates the three checkpoints on 90 evenly spaced complete starts from each of the two training blocks, 540 records across regimes, at 10–90 days. It changes no architecture/loss and reads no validation states or split-3/later archive.

Date	Type	Change
2026-07-27	Start rollout diagnosis	Submitted GPU job 291102 to execute the frozen 540-record split-1 audit. Its only purpose is to classify the objective/checkpoint-gate mismatch before any bounded successor revision; it cannot authorize inference.
2026-07-27	Complete rollout diagnosis	Job 291102 completed with exit zero in 3m58s. Every seed reproduces good day-10 slow-field skill followed by failed 10–90-day RMSE-AUC and day-90 skill on balanced training chronology. Mean day-10/AUC/day-90 persistence ratios are 0.809/2.021/4.221 for SST, 0.552/1.867/3.873 for surface pressure, and 0.532/1.844/3.835 for SSH. Apply the frozen objective/checkpoint-gate-mismatch classification; report/arrays SHA-256 values are `7efdb0726125...171374/c3c403ab81b2...5f255`.
2026-07-27	Save validation figures	Generated five project-facing PNGs from the immutable job-290738 member arrays and saved them beside the lightweight validation summary. Manifest content SHA-256 `8b24b6af0686...a0845f` binds the figures to report/array SHA-256 values `9853ea7aff21...663fb/fd84199b585e...dccc`; no sealed archive was opened.
2026-07-27	Save diagnosis figures	Generated three hash-bound project-facing figures, including a direct training-versus-validation slow-field comparison, plus a lightweight summary under `outputs/af_fno/C/rollout_diagnosis_v1/`. Manifest content SHA-256 is `664e66a0866d...393671`; no sealed archive was opened.
2026-07-27	Classify two-input/two-output option	Confirmed that current C is one-state-in/one-residual-out in time; its 51/46 figures are physical channel counts. Samudra used two consecutive states to predict two future states, whereas Bire named that method as future work. Preserve it as a bounded temporal ablation compared against the smaller long-rollout-objective correction rather than adopted without evidence.
2026-07-27	Freeze Bire-style figures	Before reading any 100–200-day validation metric, froze contract SHA-256 `791c84f12c2d...eb696b`. It fixes one seed-20260723 checkpoint, 15 randomly drawn complete S2 validation starts, the first draw for the spatial example, native-1° streamfunction truth/prediction/difference panels, and $\Delta t = 10$ RMSE curves through day 200. Persistence repeats the initial state; climatology uses all split-1 pointwise training means. The characterization cannot tune or authorize inference.
2026-07-27	Start Bire-style figures	Submitted GPU job 291134 under the unchanged contract. It reads only split-1 training states for climatology and the already opened S2 fresh-validation states for the 15 fixed rollouts; inference and all later archives remain sealed.

Date	Type	Change
2026-07-27	Complete Bire-style figures	Job 291134 completed with exit zero in 51 seconds. The fixed member's day-10/20/30/40 streamfunction RMSE is 0.0316/0.0934/0.1580/0.1765 Sv against about 13.34 Sv truth RMS. All 15 rollouts are finite, but day-200 Model-C/persistence RMSE is 0.07344/0.00480 $\mathrm{m\,s^{-1}}$ for speed, 5.981/0.0505 °C for SST, and 3.747/0.0253 $\mathrm{m^2\,s^{-2}}$ for surface pressure. The requested SST/pressure axes clip the model mean from days 100/160. Preserve report/arrays SHA-256 `b704b921444d...e300ac5/a73ac9bf2cc9...20b1` and output manifest-content SHA-256 `258be6fabe13...827f`; no inference data were opened.
2026-07-27	Freeze per-lead streamfunction maps	Before evaluating the new maps, froze contract SHA-256 `ccb965e58bdc...ae1934c`. It reuses the fixed S2 start 6335 and checkpoint, generates exactly eight separate day-20–90 truth/prediction/difference figures, fixes one shared state scale, and requires a labeled lead-specific difference scale with numerical RMSE/max error. The original output remains immutable; inference remains sealed.
2026-07-27	Complete per-lead streamfunction maps	GPU job 291135 completed with exit zero in 35 seconds and all eight requested PNGs passed hash checks. RMSE/max error grows from 0.0934/0.7240 Sv at day 20 to 0.2951/1.1713 Sv at day 90; relative RMSE remains 0.700–2.211%. Days 20–40 reproduce the original maps to FP32 roundoff. Preserve report/arrays SHA-256 `71c6e1dc1f9a...89c206/b4126494d9c0...3bcd48` and manifest-content SHA-256 `3b74d759d551...450ac7`; inference remains sealed.
2026-07-27	Save Figure-4 baseline companion	Generated a full-range/baseline-zoom SST and surface-P/ρ figure and a complete CSV from only the immutable job-291134 arrays. SST loses both baselines at day 20; pressure loses persistence at day 30 and climatology at day 70. Contract/manifest-content SHA-256 values are `2a2324090b83...b2a4a/01433936e406...99063`; no model or dataset was read.
2026-07-27	Freeze and start exact checkpoint replay	Frozen contract SHA-256 `494ce9de6688...06682` replays the original seed-20260723 optimization without changing architecture, loss, optimizer, batch order, or exposure. It saves all six late checkpoints, requires bitwise equality at the original step 14,880, and scores fixed split-1 SST/P/ρ curves against both baselines before permitting either checkpoint-only selection or one bounded rollout-objective test. Preflight, lint, shell validation, and 20 related tests pass. Submitted only this gate as GPU job 291164; no objective variant is queued.

Date	Type	Change
2026-07-28	Complete exact checkpoint replay	Job 291164 completed normally in 1h19m14s. The original selected state is bitwise exact and the six training records have zero numerical difference. All checkpoints are finite and land-zero, but none passes the long-horizon gate. Step 14,400 is diagnostic best with worst primary AUC ratio 2.366 and worst slow-field lead ratio 5.513; the result requires objective correction.
2026-07-28	Start bounded pushforward correction	Frozen contract SHA-256 `406ee5248fb4...bccc2` makes one change from step 14,400: alternating detached day-60/day-90 SST and surface-P/ρ terminal supervision with coefficient 0.0025. Width, modes, layers, padding, normalization, loss-v1 core, and split-1 gate are unchanged. Sixteen related tests and preflight pass; V100 job 291612 is running and no replica or sealed-state job is queued.
2026-07-28	Reject first push-forward gate	Job 291612 completed normally in 20m37s and selected reload-exact step 1,920. Worst primary AUC improves from 2.366 to 1.692, worst slow-field lead ratio from 5.513 to 3.994, and ten-day worst regime/group ratio from 0.852 to 0.610, but neither SST nor surface pressure beats both baselines through day 90. Publication manifest-content SHA-256 is `2b7cc78262e6...fa272`; no replication or validation.
2026-07-28	Start replay-verified duration extension	All pushforward training terms are still descending and the low learning rate covered only the final 480 steps. Frozen contract SHA-256 `4867fa77d68e...78c79` replays the complete path and requires bitwise step-1,920 equality before adding 3,840 steps at 2×10^{-5}. Nine related tests, lint, shell validation, and preflight pass; V100 job 291616 is running and all sealed reads remain blocked.
2026-07-28	Reject pushfor-ward duration	Job 291616 completed normally in 51m31s and reproduced pushforward-v1 step 1,920 bitwise. All eight checkpoints fail. Diagnostic-best step 5,760 improves worst primary AUC/slow-lead ratios to 1.597/3.609, but day-90 SST is 2.402/3.609 and surface-P/ρ 2.922/1.727 times persistence/climatology. Published manifest-content SHA-256 is `e97e78d5729c...e0f51`; replication and validation remain blocked.
2026-07-28	Correct and start three-call truncated unroll	Initial job 291769 stopped before training or metrics because the source validator expected `fine_tune_step`, while the duration checkpoint records `total_fine_tune_step`. Version-2 contract SHA-256 `0c5aabadbe51...92e3c` validates the actual schema during preflight and changes no scientific setting. Four focused tests, lint, shell validation, source hashes, and preflight pass; V100 job 291778 is running on node112.

Date	Type	Change
2026-07-28	Reject three-call truncated unroll	Job 291778 completed normally in 20m01s and diagnostic-best step 1,920 reloaded a nine-step rollout bitwise. Every checkpoint fails; worst primary AUC/slow-lead ratios 1.933/3.701 are worse than duration step 5,760 at 1.597/3.609. Day-90 SST is 2.463/3.701 and surface-P/ρ 2.844/1.681 times persistence/climatology. Published figure/CSV/summary manifest-content SHA-256 is `cccd8e757495...ae4c03`. Stop automatic variants and keep validation sealed.
2026-07-28	Freeze slow-field bias/projection audit	Contract SHA-256 `fd7d39f5c6a4...05428` fixes the seed-20260723 width-128 checkpoint, all 5,020 split-1 teacher-forced pairs per regime, the exact 540 job-291102 starts, five post-hoc correction variants, five SST/SSH increment EOFs, damped persistence, and 720-day training decorrelation. Static bias is declared dominant only if fixed 9ϵ explains at least half the pooled SST and SSH day-90 error energy; a 20% worst-slow-AUC reduction is the alternate material-correction threshold. No held state or model weight may change.
2026-07-28	Complete stream-function side study	V1 verified the project centered-U operator exactly but its 10.9% U/V comparison placed U and V on different C-grid faces. Preserved that result as a warning, froze corrected v2 SHA-256 `88ff2f20fe1a...dea170` with unchanged thresholds, and compared raw-face prefix integrals at common corner indices. S0–S2 all pass: daily p90 relative RMSE is at most 3.25×10^{-5}, annual RMSE is 4.74–6.00×10^{-5} Sv, correlations are effectively one, and boundary-normal transport is zero. Retain the free-surface caveat.
2026-07-27	Align final forward gate with Bire	Before fresh v2 validation, separate the strict all-group training-capacity test from final forward selection. Bire's surface speed, SST, and derived surface pressure use 10–90-day RMSE/ACC curve integrals against persistence/climatology; streamfunction carries circulation/stability and wind-response physics. SSH remains mandatory with bootstrap and climate-stability diagnostics but is not a standalone veto, because Bire neither predicted nor scored it separately and surface pressure already includes $g\eta$.
2026-07-28	Complete slow-field bias/projection audit	Job 292064 completed normally in 7m09s on the frozen 540 split-1 starts and opened no held archive. Fixed 9ϵ explains 33.43%/27.21% of pooled SST/SSH day-90 error energy; basin mean plus five increment EOFs explain 4.57%/1.73%. Bias plus constraints reduces worst slow-field AUC from 1.986 to 1.603 (19.25%), below the frozen 20% threshold, while conservation alone slightly worsens it. Apply the feedback-amplification branch. Report/arrays SHA-256: `e66b263c826c...0f90/4e7f73c2384e...8547`.

Date	Type	Change
2026-07-28	Freeze rollout-conditioned loss v3	Contract SHA-256 `37f1157851ac...5a08` starts at duration step 5,760 and keeps the width-128 one-state interface, loss-v1 core, data order, and 540-record gate. It applies exact SSH and regime/level temperature-mean projections after every call, adds a weight-0.01 per-cell batch-mean slow-increment penalty, and cycles a weight-0.0025 one-call forecast-age loss over days 10–90. Five focused tests, lint, shell validation, source hashes, exact forcing-regime identification, and sealed-data preflight pass; no replica or held-state job is authorized.
2026-07-28	Complete rollout-conditioned loss v3	GPU job 292293 completed normally in 27m46s without opening held states. Diagnostic step 960 reloads exactly and all ten-day groups beat persistence at U/V/temperature/SSH ratios 0.0382/0.1058/0.2746/0.3456. The strict long gate fails: worst primary AUC improves from 1.597 to 1.462, but worst slow-field lead is 3.634. SST and surface-P/ρ still reach 2.419 and 3.019 times persistence at day 90. Their day-10/day-30 values, surface speed, and streamfunction beat both persistence and climatology, so freeze the separate horizon-matched gate next; do not replicate or launch another unconstrained variant.
2026-07-28	Freeze pointwise-anomaly direct-state Model C	Contract SHA-256 `fa38b41578f5...8acea` fixes pooled S0–S2 split-1 pointwise temporal centering/scaling, a per-channel fifth-percentile wet-cell scale floor, and direct normalized future-state prediction. Width 128, modes $(24, 16)$, four layers, 10% padding, the $4C$ Channel MLP, loss v1, optimizer, and data order remain fixed. Seven checkpoints use only the exact 540 training rollouts for selection; only afterward may the same fixed 15 S2 members produce the requested day-200 figure. Eight tests, lint, shell validation, source/normalizer hashes, full preflight, and a production-grid differentiable smoke test pass. No inference, intermediate-wind, response, or adjoint state is authorized.
2026-07-28	Pass anomaly-direct reference seed	V100 job 303447 completed normally in 1h29m32s and selected step 14,400 using split 1 only. Worst training primary AUC/slow-field-lead ratios are 0.394/0.519. Fixed-S2 speed/SST/surface-P/ρ 10–90-day AUC ratios to persistence are 0.222/0.206/0.601; day-200 RMSEs are 0.00212/0.0288/0.0552 and every requested axis remains unexceeded. Save the exact requested Figure 4 plus five companion plots, CSV, summary, and manifest. Report, arrays, and checkpoint hashes are `2d840fb91682...feb6d5`, `d012c2851246...5270`, and `1d4b7cae8249...362e`.

Date	Type	Change
2026-07-28	Freeze anomaly-direct replication	Contract SHA-256 `dd7e0f92ef50...d8325f` predeclares seed-only replicas 20260724/20260725, the unchanged training-only checkpoint rule, the same fixed 15 S2 members, per-seed 10–90-day baseline/AUC and day-200 boundedness gates, and median-seed selection only if every seed passes. Seven focused tests, lint, shell validation, exact parent/result/source hashes, and both production-data preflights pass. No architecture change or later archive read is authorized.
2026-07-28	Start anomaly-direct replication	Submit V100 array 303640 with tasks 0/1 mapped immutably to seeds 20260724/20260725. Each task trains from scratch, selects among seven checkpoints on the same 540 split-1 rollouts, and only then repeats the already fixed S2 day-200 characterization and six-plot package. No inference, intermediate-wind, response, or adjoint archive is opened.
2026-07-29	Pass anomaly-direct replication	Array 303640 completed both tasks normally in 1h29m30s/1h26m32s. Every seed passes the frozen training, 10–90-day primary-AUC, day-200 boundedness, exact-reload, zero-land, and seal checks. Frozen ranking scores select median seed 20260724 at step 13,440. Publish the top-level summary, CSV, manifest, and comparison plot under `outputs/af_fno/C/anomaly_direct_replication_v1/`.
2026-07-29	Freeze held-inference forward gate	Contract SHA-256 `d11029fce3aa...c2f6c` fixes the median checkpoint, 45 starts wholly within the fresh v2 inference block, 36 ten-day calls, persistence, regime climatology, damped persistence, same-start A0, a separate 10/30-day adjoint-readiness decision, and the complete day-360 boundedness/spectra/wind-response gate. Two focused tests and source/artifact preflight pass. Intermediate-wind, response, and adjoint archives remain sealed.
2026-07-29	Start held-inference forward gate	Submit V100 job 303993 under the unchanged source- and artifact-locked contract. This is the first authorized read of the declared fresh inference states; it does not train or modify Model C and cannot open intermediate-wind, response, or adjoint archives.
2026-07-29	Complete held-inference forward gate	Job 303993 completed normally in 3m00s. Days 10/30 pass every primary RMSE/ACC comparison, all primary 10–90-day AUCs beat persistence, climatology, damped persistence, and same-start A0, normalized maximum is 15.81 with negligible group drift, and wind-slope ratio is 1.180. The combined gate fails only because day-360 mid/bottom-P/ρ modewise ratios 22.29/9.63 exceed four. Preserve the short determination as a pass but keep the complete adjoint campaign closed.

Date	Type	Change
2026-07-29	Diagnose pressure spectral failure	Freeze the original gate result and publish a separate immutable posthoc package. Total integrated mid/bottom pressure energy is 1.029/0.947 times truth; $k \geq 10$ holds only 0.0159/0.0193% of model energy, so the error is a localized small-amplitude high-wavenumber tail rather than basin-scale drift or runaway.
2026-07-29	Freeze training-only spectral attribution	Contract SHA-256 `afb64d5a0f3a...d0246` fixes all three accepted seeds, 36 split-1 starts across both training blocks, all 15 pressure levels, 36 ten-day calls, the original mode-validity rule, integrated energy, tail fractions, and onset diagnostics. Four focused tests, lint, immutable-source preflight, and record hashes pass. No validation, inference, response, or adjoint state may be read.
2026-07-29	Start training-only spectral attribution	Submit V100 job 304125. It rolls the three unchanged accepted checkpoints on the same 36 fixed split-1 starts through day 360 and writes all-level onset, modewise-ratio, integrated-energy, tail-fraction, and primary-skill guards. It changes no weight and opens no validation, inference, response, or adjoint state.
2026-07-29	Reject attribution v1 temporal alignment	Job 304125 completed in 1m07s, but the runner compared model call s with daily truth at $t + s$ instead of $t + 10s$. The 10–90-day primary ratios 3.48–4.48 times persistence are an alignment alarm, not model skill. Retain the package as rejected provenance and do not use its seed-consistency classification.
2026-07-29	Freeze corrected attribution v2	Contract SHA-256 `29c344310e50...a3de1` fixes ten daily snapshots per model call, an inclusive split-1 check through $t+360$, and 12 replacement times crossed with three regimes. It source-locks the rejected v1 package and runner, preserves the same seeds and diagnostics, and opens no held or later archive. Six focused tests, lint, shell validation, and production-data preflight pass.
2026-07-29	Start corrected attribution v2	Submit V100 job 304177. It evaluates the three unchanged accepted checkpoints on 36 fixed, fully split-1 records with truth at exact ten-day offsets through day 360, changes no weight, and opens no validation, inference, response, or adjoint state.
2026-07-29	Complete corrected attribution v2	Job 304177 completed normally in 1m11s. Every seed shows the mid/bottom factor-four tail, with day-360 ratios 12.25–13.54/4.96–5.69 and onset at days 170–190/310–330. Integrated energy remains 1.02–1.07 times truth and every primary 10–90-day ratio is 0.19–0.37 times persistence. Classify a seed-consistent localized tail and authorize exactly one bounded training-only fine-tune.

Date	Type	Change
2026-07-29	Freeze deep-pressure spectral fine-tune	Contract SHA-256 `4f3377382dc0...acef37` fixes the median source, unchanged architecture/coordinates/base loss, one 10^{-3}-weighted physical-PHIHYD modes-10–30 term over levels 7–14, 1,920 steps, and five checkpoints. Selection uses only the corrected 36 split-1 trajectories and requires factor-four spectra at every lead, bounded integrated energy, and protected primary skill. Four tests, lint, shell validation, source locks, and preflight pass; later archives stay sealed.
2026-07-29	Start deep-pressure spectral fine-tune	Submit V100 job 304324. It changes only the frozen physical-PHIHYD high-mode term, evaluates five checkpoints on the corrected split-1 day-360 protocol, and opens no validation, held-inference, response, adjoint, or long-term statistical archive.
2026-07-29	Complete deep-pressure spectral fine-tune	Job 304324 completed normally in 10m25s. The final checkpoint lowers mid/bottom day-360 ratios to 6.19/2.97, removes bottom failure and delays mid onset to day 280, while energy remains 1.006/0.980 times truth and primary skill improves. It still fails the frozen mid-level factor-four rule, so promote no checkpoint, retain original seed 20260724/step 13,440, and do not extend the observed loss weight or open later archives.
2026-07-29	Freeze Bire regularization controls	Contract SHA-256 `dd9089e9675a...0f827` fixes LayerNorm-only, dropout-0.5-only, and combined from-scratch arms with identical seed, data order, architecture apart from declared regularizers, loss, optimizer, checkpoints, and 15,360-step budget. The unregularized model is the fixed reference; selection uses only the corrected 36 split-1 rollouts. Four tests, lint, shell validation, source locks, all-arm preflight, and full-grid forward smoke tests pass.
2026-07-29	Start Bire regularization controls	Submit V100 array 304376 with task mapping 0=LayerNorm, 1=Channel-MLP dropout 0.5, and 2=combined. Each task trains from scratch with identical controlled constants, evaluates seven checkpoints only on split 1 through day 360, and opens no validation, held, long-term, response, or adjoint state.
2026-07-29	Reject Bire regularization controls	All three array tasks complete normally, but none passes. At their diagnostic checkpoints, LayerNorm/dropout/combined have mid/bottom day-360 ratios 9.74/4.92, 37.59/15.04, and 19.51/8.27; worst primary degradation is 18.6%, 57.4%, and 63.2%. Publish the source-locked aggregate whose hashes are recorded in the artifact table, retain the original checkpoint, and do not tune either regularizer.
2026-07-29	Freeze three-layer/no-padding control	Contract SHA-256 `7d405cfb8307...fd20` fixes one faithful combined Bire architecture control: three FNO blocks and no padding, with all other representation, architecture, loss, optimizer, seed, checkpoint, and corrected split-1 selection choices unchanged. Nine focused tests, lint, shell validation, source locks, production preflight, and full-grid smoke pass; later archives stay sealed.

Date	Type	Change
2026-07-29	Preserve failed architecture-control job	Job 304526 exits 1 after 46m37s, but all 15,360 training steps and all seven checkpoint writes completed. Failure occurs before the first metric when the evaluator applies an old four-layer-only reconstruction check. Assign no scientific result and preserve every checkpoint and the failed log unchanged.
2026-07-29	Freeze evaluation-only recovery	Contract SHA-256 `9232cea2ae58...165b` locks job 304526, its log, all seven checkpoint bytes, the original architecture/training contract, and zero retraining steps. It corrects only architecture reconstruction for the unchanged 36-record split-1 gate. Eleven focused tests, lint, shell validation, source locks, and production preflight pass; later archives stay sealed.
2026-07-29	Reject three-layer/no-padding control	Zero-retraining recovery job 304597 completed normally in 2m41s. No checkpoint passes. Diagnostic-best step 14,400 gives day-360 mid/bottom ratios 12.51/5.84, onset days 210/310, energy ratios 1.032/1.011, and 13.1% worst primary degradation. Publish report/manifest content SHA-256 `a429db0f64bb...58416/152190e6d06f...14539`; retain original seed 20260724/step 13,440.
2026-07-29	Freeze group-specific-head control	Contract SHA-256 `932afdd1b6b5...b1c6` fixes a shared source trunk with independent U/V/temperature/SSH projection and local heads, while retaining the representation, seed, loss, optimizer, schedule, checkpoints, and corrected 36-record split-1 gate. Three tests, lint, shell validation, source locks, production preflight, and a finite native-grid forward pass succeed; all later archives remain sealed.
2026-07-29	Reject group-specific-head control	Job 304708 completed normally in 1h30m46s. No checkpoint passes; diagnostic-best step 14,880 protects primary skill within 5.5% of source and improves source mid/bottom ratios 13.37/5.57 to 9.21/4.40, onset to days 220/350, and energy to 1.004/0.995, but still exceeds factor four. Publish report/manifest content SHA-256 `fa25da048b5d...0f3b6/8e9c6a867727...5488e`; retain the original model.
2026-07-29	Freeze S0 day-2000 truth and figure suite	Contract SHA-256 `dc204636660d...d75ea5` fixes $\tau_0 = 0.1\ \mathrm{N\,m^{-2}}$, selected seed 20260724/step 13,440, 15 RNG-selected starts from late inference indices 6660–7199, one fixed spatial member, every field/lead/baseline/filename in the six requested figures, and an exact six-year continuation from the year-120 S0 pickup. The 2,160 new daily records are evaluation-only and cannot enter training. Three focused tests, lint, shell validation, MITgcm/source preflight, and horizon coverage pass.

Date	Type	Change
2026-07-29	Complete S0 day-2000 truth	CPU job 304735 completed normally in 6m59s with scheduler exit 0. Iterations 3,110,400–3,265,920 yield all 2,160 daily dynamic/surface pairs, exact daily spacing, a final pickup, and normal termination on all four ranks. Result/manifest SHA-256 is `9ebd446ff6f3...99925/e15530c67dbe...41c6f`; preserve the isolated evaluation-only root.
2026-07-29	Freeze S0 Bire Figure 3–8 evaluation	Contract SHA-256 `6afa41cfaaf8...616030` binds job-304735 truth, both unchanged Model C checkpoints, selected pointwise normalization, all 15 starts, training-only S0 climatology, persistence, exact metric reductions, six plots, and a non-overwriting output package. Figure 6 is explicitly a prior-residual versus selected-anomaly-direct architecture-direction comparison, not a literal pretrain/fine-tune claim. Four tests, lint, shell validation, truth/source preflight, and both model smoke tests pass.
2026-07-29	Complete S0 Bire Figure 3–8 evaluation	GPU job 304736 completed normally in 1m07s and generated all six plots. At day 200 the selected model beats both baselines for speed, surface P/ρ, and SST and preserves ACC 0.784/0.887/0.972/0.833 for U/V/P/ρ/SST. At day 2,000, mean RMSE is 81.9/67.7/87.1 times persistence and the fixed member's streamfunction RMSE is 96.2 Sv; reject long-term stability while retaining the model for deterministic 10–30-day work. Publish arrays/report SHA-256 `30abd95810d3...3c7/32ac2e051726...bd18` and manifest-content SHA-256 `593dd43de772...57d1`.
2026-07-29	Freeze S0 boundary/checkpoint diagnosis	After the single-member Figure 7 shows 71.3% of day-2,000 streamfunction squared error in the four-cell boundary union, freeze contract SHA-256 `e859999279a7...d2c0` before the corresponding ensemble/checkpoint metrics. Evaluate all seven unchanged seed-20260724 checkpoints through day 2,000 on the same 15 starts, separating $Q_x/Q_y/\psi$ boundary and interior growth, normalized-amplitude crossing time, and spatial agreement. The diagnostic cannot retrain or reselect. Four tests, lint, shell validation, complete checkpoint/source/truth preflight, and all-checkpoint one-step smoke pass.
2026-07-29	Classify S0 boundary/checkpoint mechanism	Job 304740 completed normally in 1m46s and all ten manifest artifacts verify. No earlier checkpoint avoids the runaway: mean normalized amplitude 20 is crossed by days 410/580/780, versus day 880 for selected step 13,440. The four-cell boundary union contains 72.8/85.2/72.7% of selected day-2,000 $Q_x/Q_y/\psi$ squared error, and 15-member spatial cosines are 0.783/0.649/0.902. Boundary error dominates but does not consistently grow before the interior; reject checkpoint age and a simple propagation cascade, and classify a reproducible boundary-amplified global transport mode without reselecting weights.

Date	Type	Change
2026-07-29	Replace the long-stability instrument	Post-hoc contract SHA-256 `ab6d597843b1...fc5be7` confirms persistent multiplicative growth from immutable job-304736 arrays. Days-300–600 speed/P/ρ/SST gains are 1.0332/1.0397/1.0456 per call and normalized maximum gains 1.0252; all six artifacts verify. Retain 10–90-day skill separately, replace day-200 finiteness by at least 200-call saturation/statistical gates, and report singular/tangent growth rather than claiming a Jacobian spectral radius. Treat LayerNorm as an already-failed short/spectral causal control, not a presumed solution.
2026-07-29	Freeze zero-retraining stability/tangent comparison	Contract SHA-256 `5d32d472509a...bca4` binds the selected direct map, already-trained LayerNorm diagnostic, prior residual comparator, exact 15 S0 starts, job-304735 truth, 200-call statistical/spectral metrics, and three training-only tangent states. Tangent gains use shared pointwise coordinates for the direct maps only and are not spectral-radius estimates. Three tests, lint, shell validation, immutable hashes, complete-truth preflight, and finite land-zero one-step reloads of all three models pass; results remain pending and no checkpoint can be promoted.
2026-07-29	Preserve failed stability/tangent job	Job 304750 exits 1 after 1m42s. Preflight passes, but an explosive comparator overflows float32 squared-error/spectral reductions and the log-growth fit rejects the non-finite samples. The transactional writer publishes no output package; record an operational diagnostic-reduction failure, preserve failed-log SHA-256 `ae476ab3b0c...1356`, and assign no completed comparison.
2026-07-29	Freeze non-finite-safe evaluation recovery	Recovery contract SHA-256 `8bc5c0937d0e...e1ac` source-locks job 304750, the original contract, and every unchanged model/protocol input. Only physical diagnostic reductions move to float64; memberwise first non-finite lead and censored fit windows replace invalid ordinary fits. Four tests, lint, shell validation, immutable hashes, original preflight, and one-step all-model smoke pass. No retraining, selection, or promotion is permitted; results remain pending.
2026-07-29	Complete stability/tangent recovery	Job 304751 completed normally in 2m12s and all ten artifacts verify. Every state remains finite, yet the prior residual comparator reaches 10^{25}–10^{27} RMSE and is reclassified as catastrophically unstable. LayerNorm reduces selected day-2,000 speed/P/ρ/SST error by 94.9/78.7/92.1%, but remains 5.54/6.48/10.12 times climatology and has high-k power 25.5/38.9/77.0 times truth. Its high-k tangent gain is lower while full/mid ten-call gain is higher, so classify selective spectral/off-manifold stabilization rather than uniform contraction; promote no checkpoint.

Date	Type	Change
2026-07-29	Freeze training-selected high-k damping control	Contract SHA-256 `cf2f1c179d36...9554` fixes the selected checkpoint, reflected 3×3 normalized-anomaly smoother, five strengths including exact zero, ten split-1 S0 day-1,000 windows, a 5% short guard, and prospective long RMSE/amplitude/high-k gates. Only a passing nonzero strength may open the fixed S0 day-2,000 characterization. Three tests, lint, shell validation, hashes, zero-strength identity, and native-grid filtered smoke pass; zero retraining and no promotion.
2026-07-29	Reject fixed high-k damping	Job 304753 completes normally in 1m07s and its artifact package verifies. No nonzero α passes the prospective training gate. The weakest value, 0.02, reduces day-1,000 worst RMSE/climatology, amplitude, and high-k power to 0.7768/0.7742/0.4448 of source, but degrades worst short RMSE to 1.0658 times source; stronger values are worse. Keep fixed inference sealed, select no filter, promote no checkpoint, and treat high-k suppression as causal but insufficient without a state- or scale-aware learned gate.
2026-07-29	Freeze reflected spectral-gate control	Contract SHA-256 `69b0100aae27...b8e7` replaces the rejected all-scale binomial smoother by a reflected radial transfer that is identity through $k = 8$, transitions through $k = 10$, and attenuates only the diagnosed high band. The same ten split-1 S0 trajectories, 5% short guard, long gates, and conditional inference seal remain fixed. Six combined tests, lint, shell validation, source/artifact hashes, exact identity, low-mode preservation, high-mode damping, and native-grid preflight pass; no retraining or promotion.
2026-07-29	Reject reflected spectral gate; redirect forward baseline	Job 304754 completes normally in 1m06s and all nine artifacts verify. At the weakest nonzero strength, day-1,000 high-k power is only 0.0638 of source, but worst short RMSE is 6.49 times source and absolute day-1,000 amplitude/worst RMSE remain 3.65/4.41 times truth/climatology. No strength passes; keep inference sealed and promote nothing. Treat high-k excess as a symptom/amplifier, close fixed postprocessing, and test the 46-versus-10-output bottleneck with a reduced Bire-facing forward model before adjoint expansion.

Date	Type	Change
2026-07-29	Freeze Arm R reduced-channel causal control	Contract SHA-256 `a1b8ce50d1b...5a6c5` changes only the selected Model C dynamical task to ten ordered Bire-facing channels, with an explicit five-group adaptation of the unchanged three-step objective form. A transactional 2.2-GB cache (logical SHA-256 `b24728abf50f...c5aa`) reproduces four fresh full-state derivations bitwise. Seed 20260724, optimizer, 15,360 steps, seven checkpoints, 540 split-1 selection rollouts, and exact reload remain fixed. After selection, the same 15 S0 starts, day-2,000 truth, baselines, reductions, and six Figure 3–8 names run automatically. Six focused tests, lint, shell validation, source locks, both preflights, and a native-grid smoke pass; results pending and no adjoint authorization.
2026-07-30	Complete Arm R and reject the channel-count hypothesis	Arm R passes its training-only checkpoint gate and the S0 10–90-day deterministic gate, and improves days 30–200 by roughly 20% (day-90 ratios 0.803/0.811/0.861 for speed/SST/P/ρ). Day-2,000 speed and SST halve to 0.242/2.149 while surface pressure worsens 4.4-fold to 11.500, and all three stay 61.8/51.6/135.0 times climatology. Days-300–600 per-call gains are 1.0247/1.0396/1.0627 against 1.0325/1.0455/1.0401 for the retained model. Raising latent-per-output conditioning 2.78→12.8 rescales the runaway without changing its growth rate, so eliminate channel count and conditioning and return to the full 46-channel state.
2026-07-30	Correct the duplicated positional encoding	The 51 inputs already carry `longitude_normalized` and `latitude_normalized`, and `positional_embedding="grid"` appends a second `GridEmbeddingND` pair before lifting on the same unpadded 62×62 grid, so lifting sees 53 channels for 51 distinct fields against Bire's single Section-3.3.1 encoding. Disable the appended embedding for the new arm; the 512-weight reduction makes this a conditioning correction, not a capacity change, and the local 3×3 branch keeps the static position channels.
2026-07-30	Freeze single-position LayerNorm arm; correct two gate clauses	New contract returns to the full 46-channel pointwise-anomaly direct-state model and changes only the duplicated positional encoding and the Bire pointwise LayerNorm after spectral and Channel-MLP mixing. Seed, normalization, modes, width, depth, padding, optimizer, checkpoint steps, and the 540 split-1 rollouts are unchanged. The source-relative veto moves 1.1→2.0 and the deep-pressure mode-wise ratio becomes reported-not-a-veto with the integrated energy bound retained, both recorded under `gate_corrections` with the 99%-energy restriction deferred to a declared code change. A passing training gate republishes the same six Figure 3–8 names plus log-scale RMSE, per-call gain intervals, and the day-2,000 spatial standard-deviation ratio; failure advances to the declared $k = 4$–9 tangent-penalty arm with no threshold tuning.

Date	Type	Change
2026-07-30	Complete and evaluate the single-position LayerNorm arm	Job 305203 trained in 1h13m and selected step 14,880 with `arm_training_gate_passed` (worst primary 0.4716, source-relative 1.384, mode-wise 9.50, integrated 1.035/1.043). Rescoring the already-trained parent LayerNorm arm under the corrected gate shows it passing at steps 14,400 and 14,880, clearing even the original 1.1 cap at 14,880; it had failed only the deep-pressure mode-wise clause, so the gate correction rather than the architecture change produced the first pass. Job 310508 then published the same six Figure 3–8 filenames under `outputs/af_fno/C/single_position_layernorm_bire_s0_inference_v1/`: day-2,000 RMSE 0.0316/0.4155/0.4822 against 0.426/5.339/2.593, and 1,700–2,000-day per-call gain 1.0082/1.0019/1.0011 against 1.0320/1.0229/1.0241. Day-90 skill is retained at 0.29/0.22/0.53 of persistence. Surface pressure regresses at short lead and loses to persistence at day 200. Promote nothing yet; the gate ranked the arms in close to the opposite order to their long-horizon gain.
2026-07-30	Correct a mislabelled published package	The frozen figure runner hard-codes `step-13440` in its README, so job 308944's package misdescribed the model. It was withdrawn, an arm-specific README bound in, and job 310508 regenerated bitwise-identical numerical artifacts. A regression test asserts the published README names step 14,880 and that the frozen runner's own text is unchanged.
2026-07-30	Blocked: project interpreter removed	The `/sw` NFS mount was repointed to the home filesystem, `/sw/anaconda/anaconda-3.12` no longer exists, and `.venv/bin/python` is a dangling symlink; this node offers only Python 3.9/3.11 against `cpython-312` site-packages. The test suite, contract preflight, and job submission are all blocked. The new contract records `validation_status.tests_run = false` and the sbatch fails fast with an explanatory message if the interpreter is absent. Restore a Python 3.12 and repoint or rebuild `.venv` before submitting.
2026-07-30	Freeze and run the Bire-aligned full-state arm	Contract SHA-256 `62aed544703f...a71363` keeps data, grid, modes, 46-channel state, ten-day map, normalization, and the held S0 suite fixed and replaces every remaining project-specific architecture and training choice with the Bire choice: three FNO blocks, six pointwise LayerNorms, the exact `oceanfourcast.PosEmbed` sine/cosine fields appended before lifting, the two linear coordinate channels removed, no external 3×3 branch, wet-cell MSE + 0.01 MAE, one-step pretraining then two-step fine-tuning, Adam 10^{-2} at batch 8 with no clipping and declared fail-fast on non-finite loss or gradient. Nineteen focused tests, lint, source hashes, and production preflight pass; both stage checkpoints are retained and the figure suite publishes both.

Date	Type	Change
2026-07-30	Reject 10^{-2}; classify collapse to climatology	Job 311335 completed in 17m46s with no non-finite loss or gradient, so fail-fast never fired, yet training MSE settled at 0.978 against the zero-anomaly value of 1.0 and day-200 ACC was $+0.06/+0.06/+0.32/+0.09$. Day-360 mode-wise 0.95, integrated energy 0.96–1.00, and amplitude 4.87 are the best values any arm has produced, because emitting the climatological field is trivially stable; reading only the long-horizon diagnostics would have recorded this as the stability fix. Zero-contract job 311783 separated the causes on the identical code path, data, and seed: 10^{-2} stuck at one-step MSE 1.03–0.94, 5×10^{-4} reaching 0.034 against persistence 0.0185. Attribute the collapse to the learning rate, not the architecture; the paper's Table 2 pairs 0.01 with a 248×248 grid, $(64, 64)$ modes, and $C_{\text{in}}, C_{\text{out}} = 11, 10$.
2026-07-30	Complete the 5×10^{-4} one-factor control	Contract SHA-256 `b00f741562fc...2633b3` moves only the learning rate and its loader reloads the parent contract to assert every other quantity equal field by field. Job 311831 completed in 31m34s. The fine-tuned checkpoint's worst primary falls 3.466→0.952 and beats persistence on all three fields, with two-step fine-tuning alone fixing surface pressure. Held S0 day-2,000 error is flat at 3.28/4.72/3.93 times climatology from about day 500, per-call gain 1.0011/1.0002/1.0000, amplitude 7.91, all 15 members converging to one state and streamfunction still correlated 0.971 with truth. Day-200 ACC $+0.402/+0.396/+0.539/+0.226$ remains far below the incumbent. An initial submission failed in 10s because the launcher, derived from the parent by substitution, still invoked the parent module against this contract; the parent loader refused it, and a regression test now asserts each launcher names its own module and contract.
2026-07-30	Correct three protocol divergences as one bundle	Contract SHA-256 `7c3e012dae74...a9729a` sets the MAE weight to the reference 0.05 found in all four `train.py` loss functions, replaces step decay by `CosineAnnealingLR`$(3, 10^{-5})$, and selects each stage's checkpoint by lowest validation loss over a seeded 10% split-1 holdout. Dropout stays at zero deliberately. Job 314709 completed in 20m43s: the gate worst primary improves 0.9518→0.8926, but held S0 day-200 surface-U ACC falls $+0.402 \to +0.261$ and the day-2,000 tail does not move. Validation selection was a no-op, choosing exactly the earlier fixed steps, so the MAE weight and schedule moved together and cannot be separated by this arm.

Date	Type	Change
2026-07-30	Record two protocol lessons	First, for the third consecutive comparison the training-only 360-day gate and the held S0 suite ranked arms in opposite directions; stop using the gate as an S0 proxy when choosing between arms. Second, the long-rollout and attractor-mean routes are already closed on the residual lineage — pushforward jobs 291612/291616/291778 and bias-projection job 292064, whose static-bias share was 33.4%/27.2% against a 50% threshold — so any revival must first re-run the zero-retraining bias-projection instrument on a Bire-aligned checkpoint, whose failure mode is a bounded displaced attractor rather than growing amplification.
2026-07-30	Freeze and complete the loss-recovery control	Contract SHA-256 `4932821d0b46...f8c570` holds the Bire-aligned architecture, Adam 5×10^{-4}, batch 8, and the 7,680-step budget fixed and restores only the incumbent group-balanced Model C loss v1 over a three-step rollout, on the argument that Bire's channel-averaged objective gives the groups multiplicities 15:15:15:1 and leaves the free surface, circulation, pressure, and slow modes weakly constrained. The loader asserts the single change and rejects stale objective prose; all four decision branches are declared before any metric. Seventeen tests pass. Job 314770 completed in 39m00s and job 314821 published the figures: gate worst primary 0.9518→0.4532, held S0 10–90-day ratios 0.436/0.395/0.398 beating persistence on all three fields, day-200 ACC $+0.48/+0.53/+0.85/+0.42$, and a flat tail at per-call gain 1.0001/1.0008/1.0000. First predeclared branch: the architecture is useful and the objective was the problem. Day-2,000 pressure nevertheless worsens to 6.47 times climatology and the arm is not converged.